

GAME THEORY & STRATEGY

A Premium University-Level Study Companion

BOOK 1 Quiz 1 Preparation Guide

Textbook + Strategy Handbook + Complete PYQ Solutions
rolled into one self-contained resource

Coverage

Week 1: Introduction & General Principles **Week 3:** Simultaneous Moves II (Mixed)

Week 2: Simultaneous Moves I (Pure) **Week 4:** Sequential Moves

Assessment: Quiz 1 PYQs (2024 T2/T3, 2025 T2/T3) · Full Solutions · Strategy Guide
· Concepts found only in PYQs

How to Use This Book

This book is engineered to be *sufficient on its own* for scoring full marks in Quiz 1. You should be able to put away the slides and lecture videos and study only from here.

- **Part I — Complete Notes (Weeks 1–4).** Every concept is built from first principles. Read the **intuition** boxes first if you are new, then the **definitions**, then the worked examples. No prior economics or mathematics is assumed.
- **Part II — PYQ Solutions.** Every recurring Quiz-1 question is solved in extreme detail (interpretation → topic → thought process → full working → traps → final answer). The official answer keys are reproduced and independently verified.
- **Part III — Concepts Found Only in PYQs.** Ideas the exam tests but the slides barely mention (annuities, congestion/Braess routing arithmetic, IESDS by a *mixed* strategy, constant-sum structure).
- **Part IV — Quick Revision Sheets.** One-page summary, equilibrium cheat-sheet, algorithm cheat-sheet, trap list, last-minute notes.
- **Part V — Exam Strategy Guide.** How to attack each question type under time pressure.

The box colour code.

Definition

Intuition / FAQ

Why it exists

Equilibrium insight

Exam tip / Remember

Mistake / Trap / Issue .

★ How marks are actually won in Quiz 1

Roughly **four problem “families”** generate almost every question across the four papers: (1) an **auction** / **IESDS** matrix, (2) the **drug-dealer mixed-strategy** game, (3) the **sequential A/B–C/D–E/F** tree, and (4) a **congestion-routing (Braess)** problem, with occasional **Bertrand** and **Hawk–Dove** items. Master those six templates (Part II) and you have seen the whole exam.

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Part I

Complete Notes (Weeks 1–4)

Week 1 — Introduction and General Principles

1.1 From Decision Theory to Game Theory

What this whole subject is about

Ordinary decision-making asks: *given the world, what should I do?* Game theory asks a harder question: *given that the others are also thinking strategically about me, what should I do?* The twist is that your best move depends on their move, whose best move depends on *your* move. Game theory is the mathematics of untangling that circular reasoning.

1.1.1 A single decision-maker: “Eating out”

Suppose you may buy one of two items.

Item	Value (to you)	Price	Net gain = Value – Price
Cheap	1000	500	+500
Expensive	2500	3000	–500

A rational decision-maker maximises *net gain*, so they choose the **Cheap** item ($+500 > -500$). Nobody else is involved; this is pure *decision theory*.

Preferences, formally

A decision-maker has a set of *outcomes* O and a *preference relation* $\succsim$ over O , where $a \succsim b$ reads “ a is at least as good as b .” We require it to be:

- **Complete:** for any a, b , either $a \succsim b$ or $b \succsim a$ (you can always compare).
- **Transitive:** if $a \succsim b$ and $b \succsim c$ then $a \succsim c$ (no circular tastes).

A *utility function* u represents $\succsim$ if $u(a) \geq u(b) \iff a \succsim b$. “Maximising utility” just means “choosing your most-preferred available outcome.”

Ordinal vs. cardinal utility — and why it matters for games

Ordinal utility only *ranks* outcomes ($u = 1, 2, 3$ and $u = 10, 20, 30$ carry identical information). For *pure-strategy* reasoning (dominance, pure Nash) only the ranking matters, so you may rescale payoffs by any strictly increasing transformation without changing the equilibria. The moment randomness enters — mixed strategies (Week 3) or chance moves — we need **cardinal** (von Neumann–Morgenstern) utility, where *differences* are

meaningful, because we compute *expected* payoffs $\mathbb{E}[u] = \sum_k p_k u(\text{outcome}_k)$. Take-away: relabelling $2, 1 \rightarrow 200, 100$ is harmless for pure NE, but changing $2, 1 \rightarrow 2, 1.9$ *can* shift a mixed equilibrium.

1.1.2 Two decision-makers: the same problem becomes a game

Now *two* families go to dinner and **split the total bill**. Each independently orders Cheap (C) or Expensive (E); values and prices are unchanged. Because the cost is shared, *your* payoff now depends on *the other family's* order. Working out each cell (your value minus your half-share of the combined bill) gives the table below — our first **game**.

		Family 2	
		C	E
Family 1	C	500, 500	−750, 750
	E	750, −750	−500, −500

Why This Concept Exists

The moment payoffs become *interdependent*, “what should I do?” has no answer until we also model what the other will do. That interdependence — not the numbers — is what makes this game theory rather than arithmetic.

1.2 What Is a Game?

Game / Strategic interaction

A **game** is a strategic interaction in which the outcome for each agent depends not only on her own action but also on the actions chosen by others. To describe a game we specify

$$\underbrace{N}_{\text{players}}, \quad \underbrace{(A_i)_{i \in N}}_{\text{actions}}, \quad \underbrace{\text{rules / information}}_{\text{who knows/moves what, when}}, \quad \underbrace{(u_i)_{i \in N}}_{\text{payoffs}}.$$

If player 1 has actions $\{a_1, a_2, a_3\}$ and player 2 has $\{b_1, b_2\}$, the outcomes form a 3×2 table.

1.2.1 The five questions: “the rules of the game”

Every game can be pinned down by answering five questions (the “science” of the situation):

1. **Who are the players?** The decision-makers — everyone who affects payoffs.
2. **Who moves when?** The *player function* (simultaneous vs. sequential).
3. **What actions are available?** Each player’s feasible action set A_i .
4. **How much does each know when moving?** The *information*.
5. **What are the payoffs?** Numbers reflecting each player’s objectives.

And the “rules” refine these: timing (simultaneous vs. sequential), one-shot vs. repeated interaction, whether some players are better informed, and whether agreements/contracts are enforceable.

Memory hook: “PAPI-R”

Players, **A**ctions, **P**ayoffs, **I**nformation, **R**ules. If you can answer these five for any scenario, you have “modelled the game.” Quiz comprehension questions almost always begin by silently asking you to do exactly this.

1.3 A Gallery of Motivating Examples

These reappear (sometimes verbatim) in the quizzes, so meet them now.

1.3.1 Example 1 — A traffic (anti-coordination) game

		Driver 2	
		F	D
Driver 1	F	3, 3	0, 4
	D	4, 0	−1, −1

Two drivers each choose to *Defer* (F) or *Dash* (D). Mutual deferring is pleasant (3, 3); if one dashes while the other defers, the dasher gains; if both dash they crash (−1, −1). This is the structural cousin of *Chicken*.

1.3.2 Example 2 — Braess's Paradox (preview)

4000 drivers travel $A \rightarrow B$ either via C (route $A-C-B$) or via D (route $A-D-B$). Edges $A-C$ and $D-B$ are congestible with cost $x/100$ minutes (x = number of cars on that edge); edges $C-B$ and $A-D$ cost a flat 45. Each driver's payoff is the *negative* of travel time. We will show (Week 2) that adding a *new, fast* road can make *everyone slower* — the paradox the quizzes love.

1.3.3 Example 3 — Adverse selection (acquiring a firm)

Firm A (acquirer) may take over firm T (target). T 's value x is unknown to A but *known to T* ; A believes x is uniform on $\{0, 1, \dots, 100\}$ (101 equally likely values, each with probability $\frac{1}{101}$). Under A 's management T is worth $1.5x$. A bids y . If T accepts: A gets $1.5x - y$, T gets y ; if T rejects: A gets 0, T keeps x . This is a game of **incomplete information** and the seed of the “winner's curse.”

Solving the acquisition — the optimal bid is zero

T knows x and accepts iff the offer beats keeping the firm: $y \geq x$. So a bid y is accepted exactly by targets with $x \leq y$ — the *worst* ones. Conditional on acceptance, x is uniform on $\{0, 1, \dots, y\}$, so $\mathbb{E}[x \mid \text{accept}] = \frac{y}{2}$. Then

$$\mathbb{E}[A\text{'s payoff} \mid \text{accept}] = 1.5 \mathbb{E}[x \mid x \leq y] - y = 1.5 \cdot \frac{y}{2} - y = 0.75y - y = -0.25y < 0 \quad (\text{for } y > 0).$$

Every positive bid loses money in expectation; the best A can do is bid $y = 0$ (acquire nothing). **This is the winner's curse:** “winning” the auction is bad news because you only win when the asset is worth little. The lesson — *condition on the event that your offer is accepted*, not on the unconditional average $\mathbb{E}[x] = 50$ — is the single most important idea in adverse-selection problems.

Common Mistake

The seductive wrong answer: “the firm is worth $1.5 \times 50 = 75$ on average, so any bid below 75 is profitable.” This ignores that A only *wins* when T is cheap. Always compute $\mathbb{E}[\text{value} \mid \text{you win}]$, not $\mathbb{E}[\text{value}]$.

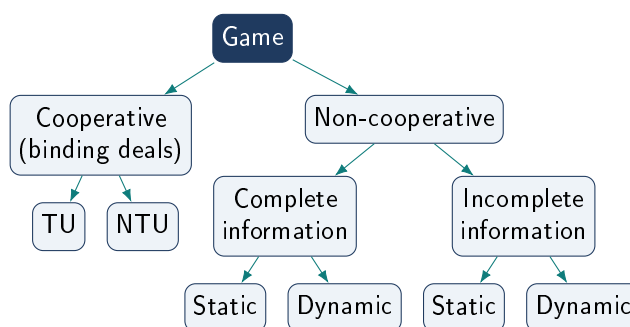
1.3.4 Example 4 — The War Game (sequential, preview of Week 4)

Country 1 decides whether to *attack* country 2 (which occupies a disputed island); if attacked, army 2 may *fight* or *retreat*. Both prefer occupying the island; a fight is worst for both. Because moves happen *in sequence*, this is naturally drawn as a **game tree** (Week 4).

1.4 Classifying Games

The map of the whole course

Almost everything you study is one branch of a single tree. Knowing where a problem sits tells you which tools apply.



Static vs. Dynamic (Simultaneous vs. Sequential).

In a *static* game players move at the same time (or in ignorance of others' moves); it is drawn as a table — the **normal/strategic form**. In a *dynamic* game players move in sequence and may observe earlier moves; it is drawn as a tree — the **extensive form**.

Complete vs. Incomplete information.

A game has *complete information* if every player knows the entire structure (players, actions, payoffs). It has *incomplete information* when at least one player is uncertain about some payoff/-type — modelled by a *probability distribution over possible values*, as in the acquisition example.

Cooperative vs. Non-cooperative.

In *cooperative* games players can form coalitions and sign *binding* agreements; we ask which coalition forms and how to split the gains (Shapley value, Book 3). In *non-cooperative* games no external enforcement exists; agreements must be *self-enforcing*. Quiz 1 is entirely non-cooperative.

★ Course road-map (what each Quiz covers)

Fundamentals: (1) Normal-form, complete info; (2) Extensive-form, complete info; (3) Normal-form, incomplete info; (4) Cooperative games. **Applications:** Matching, Voting/Cascades, Shapley value, Fair division/Bankruptcy, Auctions, Network effects. *Quiz 1 lives in fundamentals (1) and (2): normal-form and extensive-form games with complete information.*

1.5 The Prisoner's Dilemma — the most important 2×2 game

Simple Intuition

Two partners-in-crime are interrogated separately. Each is tempted to betray the other for a lighter sentence — and *individually rational* betrayal makes *both* worse off than loyal silence would. That tension between private incentive and collective good is the heartbeat of game theory.

Story (Ranga & Billa). Held in separate cells. Evidence suffices for a minor charge (1 year each) but the major charge (20 years) needs a confession. If *both confess*: 5 years each. If *one confesses*: the confessor walks free (0) and the other gets 20. If *both deny*: 1 year each. Writing payoffs as *negative* years:

		P_2	
		Confess	Deny
P_1	Confess	−5, −5	0, −20
	Deny	−20, 0	−1, −1

Formal characterisation of this game

$N = \{P_1, P_2\}$,
 Rules : choose simultaneously,
 Payoffs : $u_1(o), u_2(o)$ as in the table.

$A_1 = A_2 = \{\text{Confess, Deny}\}$,
 $O = \{(C, C), (C, D), (D, C), (D, D)\}$,

1.6 Dominance: the first solution idea

Dominant and dominated actions

Fix player i and write any action profile as (a_i, a_{-i}) , where a_{-i} collects *everyone else's* actions (the *deleted action profile*).

- Action a **strictly dominates** action b for player i if $u_i(a, a_{-i}) > u_i(b, a_{-i})$ for every a_{-i} . (We say b is *strictly dominated* by a .)
- Action a **weakly dominates** b if $u_i(a, a_{-i}) \geq u_i(b, a_{-i})$ for all a_{-i} , with strict inequality for *at least one* a_{-i} .
- a_i is a **(strictly) dominant action** if it dominates *all* other actions of player i .

The phrase “no matter what the others do” is the whole point: a dominant action is best *unconditionally*.

Apply to the PD. For P_1 : if P_2 confesses, $-5 > -20$ so Confess beats Deny; if P_2 denies, $0 > -1$ so Confess beats Deny again. Confess strictly dominates Deny — *regardless* of P_2 . By symmetry Confess is dominant for P_2 too. Hence both confess: outcome $(-5, -5)$, even though $(-1, -1)$ is better for both.

Equilibrium Insight

A *dominant-strategy* prediction is the strongest kind: a player needn't form any belief about the opponent. The PD's lesson is that rational individual choices can be collectively self-defeating — the basis of cartels collapsing, arms races, over-fishing, and price wars.

1.6.1 Pure vs. mixed actions, and domination by a mixture

A **pure** action is a deterministic choice. A **mixed** action is a probability distribution over pure actions: e.g. play U with probability p and D with $1 - p$, written $(p, 1 - p)$, with $p \in [0, 1]$ and probabilities summing to 1. Pure actions are the special cases $p = 1$ or $p = 0$.

★ A strategy can be dominated by a *mixture* even if no pure action dominates it

Consider player 1's payoffs:

		Player 2	
		L	R
Player 1	U	3	0
	M	0	3
	D	1	1

(showing only u_1). No *pure* row dominates D . But the mixed action $\frac{1}{2}U + \frac{1}{2}M$ yields expected payoff $\frac{1}{2}(3) + \frac{1}{2}(0) = 1.5$ against L and $\frac{1}{2}(0) + \frac{1}{2}(3) = 1.5$ against R — beating D 's payoff of 1 in *both* columns. So D is **strictly dominated by a mixed strategy**. This exact trick is the key to the auction/IESDS quiz question (Part II).

1.7 Equilibrium, Rationality, and Common Knowledge

Equilibrium — the guiding metaphor

An *equilibrium* is a state of balance: a state in which all forces remain in balance and which *does not change on its own*. In a game it is an outcome where, given a chance to reconsider, **no player would unilaterally want to change** their action. (For the PD, (C, C) is such a resting point.)

1.7.1 Assumption 1 — Rationality

Players are *rational*: they can compare choices (completeness), are consistent (transitivity), *maximise their payoffs*, have *perfect recall*, and are *perfect calculators*. We assume this not because humans always are, but because it gives a sharp, testable benchmark.

1.7.2 Assumption 2 — Common Knowledge

Knowledge → Mutual → Common

- **Knowledge**: “I know X .”
- **Mutual knowledge**: “everyone knows X .”
- **Common knowledge**: everyone knows X , *and* everyone knows that everyone knows X , *and* everyone knows that... — ad infinitum.

Example. If Ranga and Billa both *know* the meeting is “Monday 7pm, Rajiv Chowk metro gate 1,” and each knows the other knows, and so on, they can coordinate without further communication. The structure of the game is assumed common knowledge throughout Quiz 1.

Common beginner errors this week

- Confusing an *action* (a single move) with a *strategy* (a complete plan); they coincide only in one-shot simultaneous games.

- Checking domination against *one* opponent choice instead of *all* of them.
- Forgetting that a row can be dominated by a *mixture* of other rows even when no single row dominates it.
- Reading payoffs as costs: in the PD we use *negative years*, so “larger” means “less jail.”

End of Week 1

Quick Revision — Week 1

Game = $(N, (A_i), \text{rules/info}, (u_i))$. Describe any scenario with **PAPI-R**. **Classification**: static/dynamic, complete/incomplete, cooperative/non-cooperative. **Dominance**: a strictly dominates b if better *for all* opponent profiles; a dominant action is best unconditionally; a strategy may be dominated only by a *mixture*. **Equilibrium** = no profitable unilateral change. **Assumptions**: rationality + common knowledge.

Formula & Notation Sheet — Week 1

(a_i, a_{-i}) action profile, $a \succsim b$ preference, $u_i : O \rightarrow \mathbb{R}$ payoff.

Strict dominance: $u_i(a, a_{-i}) > u_i(b, a_{-i}) \forall a_{-i}$. **Weak dominance**: $\geq \forall a_{-i}$ and $>$ for some a_{-i} .

Mixed action of i : $\alpha_i = (p_1, \dots, p_k), p_j \geq 0, \sum_j p_j = 1$.

★ Most important Quiz points — Week 1

(i) Identify players/actions/payoffs from a word problem. (ii) Spot a strictly dominant action. (iii) Recognise domination *by a mixed strategy* (sets up IESDS). (iv) Know that complete vs. incomplete information hinges on whether *any* player is uncertain about which game is being played.

Practice questions.

1. *Conceptual*. Give a one-line scenario that is a game of *incomplete* information and explain who is uncertain about what.
2. *Numerical*. In the dinner-splitting game, verify the (C, E) cell payoff $(-750, 750)$ from first principles.
3. *Dominance*. In the 3×2 table of §1.6.1, find a mixture of U and M that gives exactly 1.5 against both columns, and explain why this makes D inadmissible.
4. *MCQ*. A game is “dominance solvable” if: (a) it has a Nash equilibrium; (b) iterated elimination of dominated actions leaves exactly one action per player; (c) every action is dominant; (d) it is zero-sum. [Ans: (b)]
5. *Proof-style*. Show that a strictly dominant action, if it exists, is unique for that player.

Week 2 — Games with Simultaneous Moves I

Where we are going

Week 1 gave us the language. Week 2 gives us the *solution concepts* for simultaneous-move (normal-form) games, in order of decreasing strength: **dominant-strategy equilibrium** → **iterated elimination** (IESDS/IEWDS) → **Nash equilibrium**. We then apply Nash to three famous models the quizzes reuse: *Braess routing*, *Hotelling location*, and *Cournot competition*.

2.1 Action versus Strategy

Action vs. strategy

An **action** is a single move. A **strategy** is a *complete contingent plan* that prescribes an action at *every* point where the player might have to move. In a one-shot simultaneous game the two coincide; in sequential games (Week 4) a strategy is a list of actions, one per decision node, so a player can have many more strategies than actions.

2.2 Solution Concept I — Dominant-Strategy Equilibrium

Strictly / weakly dominant-strategy equilibrium

Recall a_i strictly dominates a'_i if $u_i(a_i, a_{-i}) > u_i(a'_i, a_{-i})$ for all a_{-i} ; it weakly dominates if “ $\geq$ for all, $>$ for some.” A profile $a^* = (a_1^*, \dots, a_n^*)$ is a **(strictly) dominant-strategy equilibrium (DSE)** if each a_i^* is a strictly dominant action, i.e. for every i , every a'_i and every a_{-i} ,

$$u_i(a_i^*, a_{-i}) \geq u_i(a'_i, a_{-i}) \quad (\text{strict for the strict version}).$$

Prisoner’s Dilemma. Confess is strictly dominant for both, so (Confess, Confess) is the (strict) DSE.

Why a DSE is the “good news” solution

When a dominant strategy exists, a player *need not form any belief* about what others do — the best action is the same regardless. Predictions are therefore extremely robust. The catch: most games have *no* dominant strategy, which forces us to weaker concepts. Note: a game has **at most one** strictly dominant-strategy equilibrium.

Weakly dominant-strategy equilibrium

		Player 2	
		L	R
Player 1	U	1, 1	0, 0
	D	0, 0	0, 0

For Player 1: U gives (1, 0) across columns, D gives (0, 0); $U \geq D$ everywhere and $>$ when 2 plays L , so U *weakly* dominates D . Symmetrically L weakly dominates R . Hence (U, L) is a **weakly** dominant-strategy equilibrium.

2.3 Solution Concept II — Iterated Elimination

Simple Intuition

If a rational player never uses a dominated action, then a rational opponent can *delete* it, which may expose *new* dominated actions, and so on. Repeatedly deleting dominated actions is **iterated elimination**.

IESDS and IEWDS

Iterated Elimination of Strictly Dominated Strategies (IESDS): repeatedly delete strictly dominated actions until none remain. **IEWDS** does the same with *weakly* dominated actions. A game is **dominance-solvable** if iterated elimination leaves *exactly one* action for each player.

A clean IESDS

		Player 2		
		L	C	R
Player 1	U	0, 0	1, -1	1, -1
	M	-1, 1	0, 0	-1, 1
	D	-1, 1	2, 1	0, 0

For Player 1, $U = (0, 1, 1)$ vs. $M = (-1, 0, -1)$: U strictly dominates M , delete M . In what remains, for Player 2 column R becomes dominated, delete R ; then C versus L , etc. The process collapses to (U, L) .

Keynesian beauty contest — “Guess $\frac{2}{3}$ of the average”

Everyone picks a number in $[1, 100]$; the winner is closest to $\frac{2}{3}$ of the *average*. Even if all guess the max 100, $\frac{2}{3} \cdot 100 \approx 66.7$, so any guess > 67 is dominated — delete it. Now the max sensible average is 67, giving $\frac{2}{3} \cdot 67 \approx 44.3$; delete guesses above that. Iterating, the only number surviving IESDS is 1. This is the canonical demonstration that IESDS can pin down a unique outcome through layers of reasoning about reasoning.

★ The single most testable theory point this week

Iterated *strict* dominance is order-independent (path-independent): the process always terminates at the same set, and *never* deletes a Nash equilibrium. **Iterated *weak* dominance is order-dependent (path-dependent):** different deletion orders can give different survivors, and IEWDS *can* destroy a genuine Nash equilibrium. When in doubt, prefer strict elimination.

Common Mistake

Eliminating with weak dominance and then claiming a unique answer is dangerous: the answer may depend on *which* weakly dominated action you removed first. State explicitly when you use weak dominance.

IEWDS in a store-pricing game

Two stores each price High, Low, or Match (“match the rival”). Payoffs (both stores symmetric):

		Store B		
		High	Low	Match
Store A	High	10, 10	2, 15	10, 10
	Low	15, 2	5, 5	5, 5
	Match	10, 10	5, 5	10, 10

“Match” *weakly* dominates “High” for each store (it equals High against High/Match and does no worse against Low). Deleting High for both leaves a 2×2 on {Low, Match} with NE (Low, Low) = (5, 5) and (Match, Match) = (10, 10). Notice IEWDS *discarded* information: the original game also has the pure NE (High, High) = (10, 10), which weak elimination quietly removed — a concrete instance of IEWDS killing an equilibrium.

2.4 Solution Concept III — Nash Equilibrium

Simple Intuition

Most games have no dominant strategy and survive IESDS with many actions. We need a concept that always says something. Nash’s idea: an outcome is stable if *each player is simultaneously best-responding to the others*. Nobody, looking only at their own payoff, regrets their choice given what everyone else did.

2.4.1 Best response

Best-response correspondence

Player i ’s **best response** to others’ profile a_{-i} is the set of actions maximising her payoff:

$$BR_i(a_{-i}) = \arg \max_{a_i \in A_i} u_i(a_i, a_{-i}).$$

It is a *correspondence* (set-valued) because ties are possible.

2.4.2 Nash equilibrium, defined three equivalent ways

Nash equilibrium (NE)

A profile $a^* = (a_1^*, \dots, a_n^*)$ is a **Nash equilibrium** if any of these (equivalent) statements holds:

1. For each player i and each $a_i \in A_i$: $u_i(a_i^*, a_{-i}^*) \geq u_i(a_i, a_{-i}^*)$.
2. Each player is best-responding: $a_i^* \in \text{BR}_i(a_{-i}^*)$ for all i .
3. *No player can become better off by deviating unilaterally* (everyone else held fixed).

DSE $\Rightarrow$ NE, but not conversely

Every dominant-strategy equilibrium is a Nash equilibrium. The converse fails: a coordination game like

		Player 2	
		A	R
Player 1	A	3, 1	0, 0
	R	0, 0	1, 3

has two pure NE, (A, A) and (R, R) , yet *neither* action is dominant. Also: a NE is never deleted by IESDS; if IESDS leaves a unique profile it *is* the unique NE.

2.4.3 The best-response (underline) method

★ How to find all pure Nash equilibria of a matrix — fast

For each column, underline the row player's best payoff(s); for each row, underline the column player's best payoff(s). A cell with *both* numbers underlined is a Nash equilibrium. This is the method to use under time pressure.

Best response as set-valued: a 3×2 illustration

Player 1 has rows U, M, D ; player 2 has columns L, R . Showing only u_1 :

	L	R
U	8	4
M	3	6
D	5	4

$$\text{BR}_1(L) = \{U\} \text{ (8)}, \quad \text{BR}_1(R) = \{M\} \text{ (6)}.$$

If instead $u_1(U, R) = u_1(D, R) = 6$, then $\text{BR}_1(R) = \{U, D\}$ — two actions tie, which is exactly why the best response is a *correspondence*, not a function. Ties like this are the seedbed of *mixed* equilibria (Week 3).

2.5 What Does a Nash Equilibrium *Mean*? (Interpretations)

Simple Intuition

The definition tells you how to *check* an equilibrium, not *why* players would ever land on one. Several “stories” (Osborne–Rubinstein call them *interpretations*; Dutta calls them *parables*) justify the concept. Knowing them answers the common exam/interview question “but how do players reach a Nash equilibrium?”

Two families of justification (Osborne & Rubinstein)

- **Deductive / educative:** players are hyper-rational and, by reasoning about each other's reasoning, *deduce* the equilibrium before play. Strong cognitive demands.
- **Evolutionary:** the equilibrium is a *steady state* reached by adaptation, imitation, or natural selection over many rounds — no player need understand the game; bad strategies simply die out.

Nash's five parables (Prajit Dutta)

1. **Play prescription:** a trusted advisor (a textbook, a consultant, convention) recommends a profile; it is *self-enforcing* only if it is a Nash equilibrium (no one wants to disobey).
2. **Pre-play communication:** players talk first and reach a non-binding agreement; the only agreements that survive once everyone goes home are Nash equilibria.
3. **Rational introspection:** each player simulates the others' reasoning "in their head" and best-responds to the conclusion.
4. **Focal point** (Schelling): culture/context makes one equilibrium salient ("meet at the clock tower at noon"), selecting among multiple equilibria.
5. **Trial and error:** repeated play with small adjustments converges to a resting point — the evolutionary story in miniature.

The common thread: in every story the *only* self-consistent resting point is a profile where no one wants to deviate — i.e. a Nash equilibrium.

★ Exam Tip

If asked "why should we expect Nash play?", name at least one deductive justification (rational introspection) *and* one evolutionary one (trial-and-error / steady state). For *equilibrium selection* when there are several NE, invoke focal points or payoff/risk dominance (Stag Hunt).

2.6 A Catalogue of Classic 2×2 Games

These structures are tested directly

The quizzes lift these games (especially Hawk–Dove and anti-coordination "matching" games) almost verbatim. Learn to recognise the *pattern* of payoffs, not just the story.

2.6.1 Battle of the Sexes (coordination with conflicting preferences)

		Player 2	
		<i>M</i>	<i>C</i>
Player 1	<i>M</i>	2, 1	0, 0
	<i>C</i>	0, 0	1, 2

Both want to be *together* but disagree on *where*. Two pure NE on the diagonal, (M, M) and (C, C) ; a third *mixed* NE appears in Week 3.

2.6.2 Stag Hunt (assurance / coordination with risk)

		Hunter 2	
		Stag	Rabbit
Hunter 1	Stag	8, 8	0, 4
	Rabbit	4, 0	2, 2

Cooperating on the stag is best *if* the other cooperates; hunting rabbit is safe. Two pure NE: (S, S) is *payoff-dominant*, (R, R) is *risk-dominant*. Rousseau's social-contract parable.

2.6.3 Hawk–Dove / Chicken (anti-coordination)

Simple Intuition

Two animals contest a resource of value v . Each plays **Hawk** (fight) or **Dove** (display). Two Hawks fight and split v minus injury cost; two Doves split v minus a small display cost; a Hawk meeting a Dove takes all of v . The defining feature: **you want to do the opposite of your opponent** — both pure NE are off the diagonal.

With value v , injury cost k (assume $k > v/2$) and display cost d (assume $d < v/2$):

		Player 2	
		Hawk	Dove
Player 1	Hawk	$\frac{v}{2} - k, \frac{v}{2} - k$	$v, 0$
	Dove	$0, v$	$\frac{v}{2} - d, \frac{v}{2} - d$

With $v = 10, k = 6, d = 4$ this is $(-1, -1), (10, 0), (0, 10), (1, 1)$. The two pure NE are the *asymmetric* ones (H, D) and (D, H) ; (H, H) and (D, D) are *not* equilibria. (*This is exactly the 2024 T2 Hawk–Dove question.*)

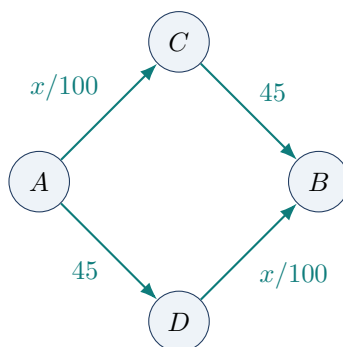
2.7 Application A — Braess's Paradox (Congestion Routing)

Simple Intuition

Adding capacity to a network can make *everyone* worse off, because self-interested routing ignores the congestion each driver imposes on others. The quizzes test this as the “choosing a route” problem.

2.7.1 The base network

4000 drivers go $A \rightarrow B$. Routes $A-C-B$ and $A-D-B$. Congestible edges $A-C$ and $D-B$ cost $x/100$ (with x cars on them); flat edges $C-B$ and $A-D$ cost 45.



Model. Players $\{1, \dots, 4000\}$; each chooses $\{ACB, ADB\}$; payoff = $-\text{travel time}$. If n take ACB and $4000 - n$ take ADB :

$$T_{ACB} = \frac{n}{100} + 45, \quad T_{ADB} = 45 + \frac{4000 - n}{100}.$$

At a Nash equilibrium no driver can switch and gain, which forces $T_{ACB} = T_{ADB}$, i.e. $n = 4000 - n \Rightarrow n = 2000$. Each route carries 2000 cars; every driver's time is $\frac{2000}{100} + 45 = \mathbf{65}$ minutes.

2.7.2 Add a fast road $C \rightarrow D$ and watch it get worse

Now a short link lets drivers do $A-C-D-B$, using the two *cheap* congestible edges $A-C$ and $D-B$. Every driver now finds $ACDB$ tempting. In the new equilibrium *all* 4000 pile onto $A-C$ and $D-B$:

$$T_{ACDB} = \frac{4000}{100} + 0 + \frac{4000}{100} = 40 + 40 = \mathbf{80} \text{ minutes.}$$

Everyone is *slower* (80 vs. 65) despite a new road — **Braess's Paradox**.

Real cases

Stuttgart 1969 (traffic improved only after a new road was *closed*); New York 1990 (closing 42nd Street improved flow); Seoul 2002 (tearing down a 6-lane road improved flow). Same logic appears in electricity grids and internet routing.

★ How the quiz version differs

The PYQ uses *four* drivers and *asymmetric* edge costs read off the figure: $A-X$ and $Y-B$ cost 6, 9, 12, 15 for 1–4 cars (i.e. $6 + 3(m-1)$); $A-Y$ costs 20, 21, 22, 23; $X-B$ costs 20, 20.9, 21.8, 22.7; the added $X-Y$ link costs 7, 8, 9, 10. The base NE is two per route with times 29.9 and 30 minutes; after the link the NE is (1, 2, 1) across $A-X-B$, $A-X-Y-B$, $A-Y-B$ with every time = 32 minutes (worse). Full arithmetic and a note on a misprinted “36” option are in Part II, Family 5.

2.8 Application B — Hotelling's Location Model

Simple Intuition

Two ice-cream sellers on a 1 km beach with customers spread evenly choose where to park. Each wants the larger share; customers walk to the nearer cart. Where do the carts end up?

Setup. Linear “beach” $[0, 1]$, customers uniform, each buys one; walking 100 m costs disutility 1; price fixed at 10. Sellers A, B pick locations to maximise market share.

Principle of Minimum Differentiation

A seller's best response is always to edge *just past the midpoint between herself and the rival* toward the larger segment. The only mutual best response is for *both to locate at the centre* $x_A = x_B = 0.5$: the unique Nash equilibrium. This “median” logic explains why competitors cluster (rival petrol stations, similar TV schedules) and why two parties converge to the *median voter*.

Why the centre, step by step

Customers go to the nearer cart, so if A sits at $x_A < x_B$, the *indifferent* customer is at the midpoint $m = \frac{x_A + x_B}{2}$: everyone left of m buys from A , everyone right from B . A 's market share is therefore m , which A increases by moving *right* (raising m) — right up until x_A just reaches x_B . The same pull acts on B from the other side. The only place neither wants to move is $x_A = x_B = 0.5$, where each gets exactly half. **Edges are not equilibria:** if both sat at 0, either could jump to 0^+ and capture nearly the whole line. **Caveat:** with *three* sellers no pure equilibrium exists (the middle seller is always squeezed), and adding price competition can reverse the result toward *maximum* differentiation — a favourite “limitations” discussion point.

2.9 Application C — Cournot Duopoly (quantity competition)

Simple Intuition

Two firms simultaneously choose *how much to produce*; the market price falls as total output rises. Each firm trades off “sell more” against “don't crash the price.” Cournot is the workhorse model of imperfect competition and a perfect vehicle for computing a Nash equilibrium with *continuous* actions.

Setup. Inverse demand $P = a - bQ$ with $a, b > 0$ and total output $Q = Q_1 + Q_2$. Firm i has constant marginal cost c_i and chooses $Q_i \geq 0$. Profit:

$$\pi_i = \underbrace{P Q_i}_{\text{revenue}} - \underbrace{c_i Q_i}_{\text{cost}} = [a - b(Q_1 + Q_2)] Q_i - c_i Q_i = [a - c_i - b(Q_1 + Q_2)] Q_i.$$

2.9.1 Best-response (reaction) functions

Holding the rival's Q_j fixed, maximise over Q_i :

$$\frac{\partial \pi_i}{\partial Q_i} = a - c_i - b(Q_i + Q_j) - bQ_i = 0 \implies \boxed{\text{BR}_i(Q_j) = \frac{a - c_i}{2b} - \frac{Q_j}{2}} \quad (\text{when positive}).$$

2.9.2 Equilibrium

Solve the two reaction functions simultaneously. With *equal* costs $c_1 = c_2 = c$, symmetry gives $Q_1^* = Q_2^* = Q^*$:

$$Q^* = \frac{a - c}{2b} - \frac{Q^*}{2} \implies \frac{3}{2}Q^* = \frac{a - c}{2b} \implies \boxed{Q_1^* = Q_2^* = \frac{a - c}{3b}}, \quad Q_{\text{tot}}^* = \frac{2(a - c)}{3b}.$$

Cournot at a glance

$$\text{BR}_i(Q_j) = \frac{a - c_i}{2b} - \frac{Q_j}{2}, \quad Q_i^* = \frac{a - c}{3b} \text{ (symmetric)}, \quad P^* = a - bQ_{\text{tot}}^* = \frac{a + 2c}{3}.$$

With *asymmetric costs* ($c_1 \neq c_2$), the cheaper firm produces more; solve the two linear reaction equations directly.

A full Cournot computation and three benchmarks

Let $a = 120, b = 1, c = 0$. **Cournot:** $Q_i^* = \frac{120}{3} = 40$ each, total 80, price $P^* = 120 - 80 = 40$, each profit $= 40 \times 40 = 1600$ (industry 3200).

- **Monopoly** (one firm maximises $(120 - Q)Q$): $Q^M = 60, P^M = 60$, profit 3600 — *less output, higher price, higher industry profit* than Cournot.
- **Perfect competition** ($P = c = 0$): $Q^C = 120, P^C = 0$, profit 0 — *most output, lowest price*.
- **Cournot sits in between:** total output 80 lies between monopoly's 60 and competition's 120; price 40 lies between 60 and 0. More firms $\Rightarrow$ output rises toward the competitive level.

Asymmetric costs: solving two reaction lines

$a = 100, b = 1, c_1 = 10, c_2 = 20$. Reaction functions $Q_1 = \frac{90}{2} - \frac{Q_2}{2} = 45 - \frac{Q_2}{2}$ and $Q_2 = 40 - \frac{Q_1}{2}$. Substitute: $Q_1 = 45 - \frac{1}{2}(40 - \frac{Q_1}{2}) = 25 + \frac{Q_1}{4} \Rightarrow \frac{3}{4}Q_1 = 25 \Rightarrow Q_1^* = \frac{100}{3} \approx 33.3$, then $Q_2^* = 40 - \frac{50}{3} \approx 23.3$. The *lower-cost* firm 1 produces more — the general lesson with asymmetric costs.

Common Mistake

- (i) Forgetting the “ $-bQ_i$ ” term from differentiating PQ_i (price itself depends on Q_i). (ii) Using the monopoly formula. (iii) Ignoring the non-negativity corner: if a rival floods the market, $BR_i = 0$.

End of Week 2

Quick Revision — Week 2

Strength ladder: DSE $\Rightarrow$ survives IESDS $\Rightarrow$ NE. **IESDS** is path-independent and preserves NE; **IEWDS** is path-dependent and may kill a NE. **NE** = mutual best response = no profitable unilateral deviation; find it by *underlining best responses*. **Classic games:** PD (dominance), BoS & Stag Hunt (two coordinating NE), Hawk–Dove (two *anti-coordinating* NE). **Applications:** Braess (equalise route times; new road can hurt), Hotelling (both to centre 0.5), Cournot ($Q^* = (a - c)/3b$).

Formula Sheet — Week 2

$$BR_i(a_{-i}) = \arg \max_{a_i} u_i(a_i, a_{-i}); \quad \text{NE} : a_i^* \in BR_i(a_{-i}^*) \quad \forall i.$$

Braess base: $T = \frac{n}{100} + 45$, equalise $\Rightarrow$ split evenly. **Cournot:** $BR_i = \frac{a - c_i}{2b} - \frac{Q_j}{2}$,
 $Q^* = \frac{a - c}{3b}$. **Hotelling NE:** $x_A^* = x_B^* = 0.5$.

★ Algorithm sheet

Find pure NE (matrix): underline each player's best payoff in every row/column $\rightarrow$ doubly-underlined cells are the NE. **IESDS:** delete a strictly dominated action (possibly

using a *mixture*); repeat. **Cournot:** write $\pi_i \rightarrow$ set $\partial\pi_i/\partial Q_i = 0 \rightarrow$ get $BR_i \rightarrow$ solve simultaneously. **Routing:** guess split $\rightarrow$ equalise route times $\rightarrow$ solve for split $\rightarrow$ plug back for time.

Practice questions.

1. *Best response.* For Stag Hunt, list $BR_1(\text{Stag})$ and $BR_1(\text{Rabbit})$ and confirm both diagonal cells are NE.
2. *IESDS vs IEWDS.* Construct a 2×2 game where IEWDS deletes a genuine Nash equilibrium. (Hint: ties.)
3. *Cournot numeric.* $a = 100, b = 1, c = 10$. Find Q_i^*, P^* , and each firm's profit.
4. *Braess.* In the base network with 4000 drivers, what happens to equilibrium time if a third route with constant cost 50 is added? Is it used?
5. *MCQ.* The Hotelling NE on $[0, 1]$ is: (a) $(0, 1)$; (b) $(0.25, 0.75)$; (c) $(0.5, 0.5)$; (d) no NE.
[Ans: (c)]

Week 3 — Games with Simultaneous Moves II (Mixed Strategies)

The problem mixed strategies solve

Some games — Matching Pennies, Rock–Paper–Scissors, the drug-dealer game — have *no* pure Nash equilibrium: whatever you do predictably, the opponent exploits it. The fix is to be *unpredictable*: randomise. A mixed strategy turns “what will I play?” into “with what probabilities will I play each option?” and restores the existence of equilibrium.

3.1 Why Pure Equilibria Can Fail

Zero-sum / constant-sum game

A two-player game is **zero-sum** if the payoffs in every cell sum to 0 (one’s gain is the other’s loss); **constant-sum** if they sum to the same constant k in every cell. Pure conflict games of this type frequently have no pure NE.

3.1.1 Matching Pennies

		Player 2	
		Head	Tail
Player 1	Head	1, −1	−1, 1
	Tail	−1, 1	1, −1

Player 1 wants to *match*; Player 2 wants to *mismatch*. Best responses cycle endlessly $\Rightarrow$ **no pure NE**. Rock–Paper–Scissors is the 3×3 version with the same feature.

Rock–Paper–Scissors: the symmetric mixed NE

Payoffs +1 win, −1 lose, 0 tie (zero-sum). By symmetry guess each player mixes $(\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$. Against a uniform opponent, *every* pure action yields the same expected payoff $\frac{1}{3}(0) + \frac{1}{3}(+1) + \frac{1}{3}(-1) = 0$, so each player is indifferent — the support condition holds and $(\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$ for both is the unique NE. Any deviation (say favouring Rock) is immediately exploited (opponent shifts to Paper), confirming uniform play is the only resting point. This is the cleanest illustration that randomising to make the opponent *indifferent* is what sustains a mixed equilibrium.

3.2 Mixed Strategies

Mixed strategy

A **mixed strategy** α_i for player i is a probability distribution over her pure actions $A_i = \{a_{i1}, \dots, a_{ik}\}$: a vector $(p_1, \dots, p_k)$ with $p_j \geq 0$ and $\sum_j p_j = 1$. The set of pure actions played with positive probability is the **support**. Pure strategies are the degenerate mixtures ($p_j = 1$ for one j).

Why This Concept Exists

Allowing randomisation “convexifies” the strategy space: instead of a finite menu of corners we get a continuous set. This (i) creates new equilibrium candidates and (ii) guarantees existence (Nash’s theorem). The cost is interpretation — what does it *mean* to “play Head with probability $\frac{1}{2}$ ”? (See §3.9.)

3.3 Deriving a Mixed Nash Equilibrium — the Indifference Principle

The one idea that solves every mixed-strategy question

If you are willing to randomise between two actions, those two actions must give you the *same* expected payoff — otherwise you would put all your probability on the better one. So at a mixed equilibrium, **each player’s randomisation makes the other player exactly indifferent**. You solve for *my* probabilities by setting the *opponent’s* two expected payoffs equal.

3.3.1 Matching Pennies, fully solved

Let Player 1 play Head with probability p (Tail with $1 - p$), and Player 2 play Head with probability q (Tail with $1 - q$).

Player 1’s expected payoffs (Player 2 mixes with q):

$$\mathbb{E}u_1(\text{Head}) = q(1) + (1 - q)(-1) = 2q - 1, \quad \mathbb{E}u_1(\text{Tail}) = q(-1) + (1 - q)(1) = 1 - 2q.$$

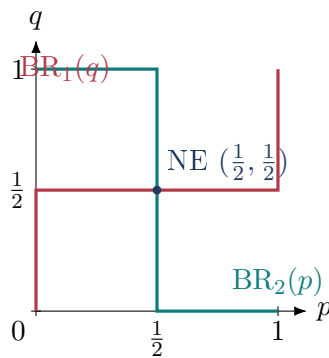
Player 1 strictly prefers Head if $2q - 1 > 1 - 2q \Leftrightarrow q > \frac{1}{2}$, prefers Tail if $q < \frac{1}{2}$, and is **indifferent** iff $q = \frac{1}{2}$. Hence Player 2’s equilibrium mix is $q^* = \frac{1}{2}$. By the symmetric computation on Player 2 (whose payoffs from Head/Tail are $1 - 2p$ and $2p - 1$), $p^* = \frac{1}{2}$.

$$\text{Mixed NE: } p^* = q^* = \frac{1}{2}.$$

3.3.2 The best-response correspondences (graphical view)

$$\text{BR}_1(q) = \begin{cases} \{p = 1\}, & q > \frac{1}{2} \\ [0, 1], & q = \frac{1}{2} \\ \{p = 0\}, & q < \frac{1}{2} \end{cases} \quad \text{BR}_2(p) = \begin{cases} \{q = 0\}, & p > \frac{1}{2} \\ [0, 1], & p = \frac{1}{2} \\ \{q = 1\}, & p < \frac{1}{2}. \end{cases}$$

Plotting these “staircases” on the unit square, they intersect only at $(p, q) = (\frac{1}{2}, \frac{1}{2})$ — the unique (mixed) NE.



★ The mechanical recipe for ANY 2×2 mixed NE

To find your opponent's mixing probability, set *your* two pure-action expected payoffs equal and solve — but remember the cross-over: *Player 1's* indifference equation involves *Player 2's* probability q , and vice versa. This is precisely how the drug-dealer question is cracked (Part II).

3.4 Existence: Nash's Theorem

Theorem 3.1 (Nash, 1950). *Every strategic game with finitely many players, each having finitely many pure strategies, possesses at least one (possibly mixed) Nash equilibrium.*

Equilibrium Insight

The theorem is *reassuring* (a solution always exists) but *not constructive* (it doesn't tell you how to find it). Finiteness is sufficient, not necessary — Cournot has infinite strategies yet a NE.

3.5 Characterising Mixed Equilibria (support conditions)

Indifference / support conditions

Let α^* be a NE and let i mix. Then:

1. **(Equal-payoff)** Every pure action *in the support* of α_i^* yields the same expected payoff: $v_i(x, \alpha_{-i}^*) = v_i(y, \alpha_{-i}^*)$ for all x, y played with positive probability.
2. **(No-better-deviation)** Every pure action *not* in the support yields no more: $v_i(z, \alpha_{-i}^*) \leq v_i(x, \alpha_{-i}^*)$.

Formally, $U_i(\alpha^*) \geq U_i(a_i, \alpha_{-i}^*)$ for every pure a_i , where U_i is the expected payoff.

3.6 Mixed Nash in the Battle of the Sexes

		Player 2	
		A	R
Player 1	A	3, 1	0, 0
	R	0, 0	1, 3

Let Player 1 play A with probability p , Player 2 play A with probability q .

Player 1 indifferent (between A and R): $\mathbb{E}u_1(A) = 3q$, $\mathbb{E}u_1(R) = 1 - q$. Set equal: $3q = 1 - q \Rightarrow q^* = \frac{1}{4}$.

Player 2 indifferent: $\mathbb{E}u_2(A) = p$, $\mathbb{E}u_2(R) = 3(1 - p)$. Set equal: $p = 3(1 - p) \Rightarrow p^* = \frac{3}{4}$.

Three NE: (A, A) , (R, R) , and mixed $(p^* = \frac{3}{4}, q^* = \frac{1}{4})$.

Common Mistake

Do *not* pair “my” probability with “my” indifference equation. Player 1’s mix (p) is pinned down by *Player 2’s* indifference; Player 1’s own indifference pins down q . Mixing these up is the #1 source of wrong probabilities.

3.7 Application — Reporting a Crime (the Bystander Effect)

Simple Intuition

A crime is witnessed by n people. Everyone wants the police called but would rather *someone else* make the call (calling has a small personal cost). As the crowd grows, does help become *more* or *less* likely? Game theory gives the unsettling answer with *constant* payoffs — no need to assume people care less in crowds.

Setup. Players $I = \{1, \dots, n\}$, actions $\{\text{Call } (R), \text{Don't } (N)\}$. If the crime is reported (someone calls), each non-caller gets benefit v ; a caller pays cost c and nets $v - c$; if nobody calls, everyone gets 0. Assume $v > c > 0$. For $n = 2$:

		Player 2	
		N	R
Player 1	N	0, 0	$v, v - c$
	R	$v - c, v$	$v - c, v - c$

Pure NE: (R, N) and (N, R) — “exactly one person calls.” There is also a *symmetric mixed* NE.

3.7.1 Symmetric mixed equilibrium with n players

Let each person call with probability p . Person i is indifferent between calling and not. Calling gives $v - c$ for sure. Not calling gives v only if *at least one of the other $n - 1$ calls*:

$$\mathbb{E}u_i(N) = v[1 - (1 - p)^{n-1}].$$

Indifference $v - c = v[1 - (1 - p)^{n-1}]$ gives

$$v(1 - p)^{n-1} = c \Rightarrow (1 - p)^{n-1} = \frac{c}{v} \Rightarrow p^* = 1 - \left(\frac{c}{v}\right)^{1/(n-1)}.$$

Diffusion of responsibility, derived

As $n \uparrow$, the exponent $\frac{1}{n-1} \downarrow 0$, so $(c/v)^{1/(n-1)} \uparrow 1$ and each individual’s call probability $p^* \downarrow$. Worse, the probability that *nobody* calls is

$$(1 - p^*)^n = \left(\frac{c}{v}\right)^{n/(n-1)} \xrightarrow{n \rightarrow \infty} \frac{c}{v},$$

which *rises* with n over the relevant range. More witnesses $\Rightarrow$ each less likely to act *and* a higher chance nobody helps — the Kitty Genovese “bystander effect,” explained with fixed benefits and costs.

3.8 Never-a-Best-Response = Dominated by a Mixed Strategy

Belief and never-best-response

A **belief** of player i is a probability distribution over the others' actions. An action is a **never-best-response** if it is not a best response to *any* belief.

Theorem 3.2 (Key equivalence). *In a finite game, an action is a **never-best-response** if and only if it is **strictly dominated by some mixed strategy**. Consequently a never-best-response is played with probability zero in every (mixed) Nash equilibrium.*

When a pure action is / isn't dominated by a mixture

Take player 1's payoffs with rows U, M, D and columns L, R :

Case A: $U = (10, 0)$, $M = (4, 4)$, $D = (0, 10)$.

Is M dominated by $pU + (1-p)D$? That mixture pays $10p$ vs. L and $10(1-p)$ vs. R . To beat M we need $10p > 4$ and $10(1-p) > 4$, i.e. $p > 0.4$ and $p < 0.6$. Achievable (e.g. $p = \frac{1}{2}$ gives 5, 5): M is **dominated** — never a best response.

Case B: $U = (10, 0)$, $M = (6, 6)$, $D = (0, 10)$.

Now we would need $10p > 6$ and $10(1-p) > 6$, i.e. $p > 0.6$ and $p < 0.4$ — *impossible*. M is **not dominated**; it is a best response to some belief (around $q = \frac{1}{2}$).

The British spy (Joel Watson, Ch. 11): finding never-best-responses

Spy 001 picks a ski route a, b, c, d ; rival 002 chooses to set off an explosive (y) or not (x). Spy 001's payoffs (rows a, b, c, d ; columns x, y) are

$$a = (12, 0), \quad b = (11, 1), \quad c = (10, 2), \quad d = (9, 6).$$

Best responses: if 002 plays x (no explosive), faster is better, so $BR = a$ (12). If 002 plays y , d is safest, $BR = d$ (6). So a and d are best responses to *some* belief. Are b, c ever best responses? Check whether $b = (11, 1)$ is beaten by a mix of a and d : the mixture $\alpha a + (1-\alpha)d = (9+3\alpha, 6-6\alpha)$. To dominate $(11, 1)$ we need $9+3\alpha > 11$ (i.e. $\alpha > \frac{2}{3}$) and $6-6\alpha > 1$ (i.e. $\alpha < \frac{5}{6}$); the window $\alpha \in (\frac{2}{3}, \frac{5}{6})$ is non-empty, so b is **strictly dominated by a mixture** — **never a best response**. A similar check disposes of c . Hence only a and d survive, and any mixed NE mixes 001 over $\{a, d\}$. *Moral:* screen out never-best-responses (via the mixture test) *before* hunting for the mixed equilibrium — it shrinks the problem.

3.9 Interpreting a Mixed Equilibrium

Simple Intuition

"Playing $\frac{1}{2}-\frac{1}{2}$ " can mean several things: (i) a single player literally randomises (flips a coin); (ii) a *population* interpretation — half the population plays Head, half plays Tail, and the proportions are the "probabilities"; (iii) a *steady state* of a large recurring interaction. Harsanyi's *purification* adds (iv): the mix approximates pure play in a nearby game with tiny private payoff perturbations.

★ Stability footnote

In *anti-coordination* games (Hawk–Dove/Chicken) the mixed equilibrium is typically *unstable* under best-response dynamics: small perturbations push play toward one of the pure equilibria. In *matching* (pure-conflict) games the mixed equilibrium is the stable resting point. Useful intuition, occasionally asked as a concept MCQ.

End of Week 3

Feature	Pure strategy	Mixed strategy
What it is	A single deterministic action	A probability distribution over actions
Exists in every finite game?	Pure NE may <i>not</i> exist	A (mixed) NE <i>always</i> exists (Nash)
Found by	Underline best responses	Indifference principle (opponent indifferent)
Typical home	Coordination, dominance games	Pure-conflict / constant-sum games
Payoff comparison	Realised payoff	<i>Expected</i> payoff $U_i(\alpha)$

Quick Revision — Week 3

Mixed strategy = distribution over actions. **Indifference principle**: your mix makes the *opponent* indifferent; solve by equating the opponent's pure-action expected payoffs (cross-over!). **Existence** (Nash): finite games always have a (mixed) NE. **Support conditions**: actions in the support tie; actions outside pay $\leq$. **BoS mixed NE** $(\frac{3}{4}, \frac{1}{4})$. **Crime reporting**: $p^* = 1 - (c/v)^{1/(n-1)}$ falls with n . **Never-best-response** $\Leftrightarrow$ **dominated by a mixture**.

Formula Sheet — Week 3

$$\mathbb{E}u_i(\text{action}) = \sum(\text{prob of opp. profile}) \times (\text{payoff}), \quad U_i(\alpha^*) \geq U_i(a_i, \alpha_{-i}^*) \quad \forall a_i.$$

2×2 mixed NE: set $\mathbb{E}u_1(\text{row}_1) = \mathbb{E}u_1(\text{row}_2)$ to get q^* ; set $\mathbb{E}u_2(\text{col}_1) = \mathbb{E}u_2(\text{col}_2)$ to get p^* . **Bystander**: $p^* = 1 - (c/v)^{1/(n-1)}$, $\Pr[\text{no one calls}] = (c/v)^{n/(n-1)}$.

★ Algorithm sheet — mixed NE

- (1) Check for pure NE first (underline method).
- (2) If asked for mixing: assign p to row player's first action, q to column player's first.
- (3) Write each player's two expected payoffs.
- (4) Equate *Player 1's* two payoffs $\rightarrow$ solve for q ; equate *Player 2's* $\rightarrow$ solve for p .
- (5) Sanity-check $p, q \in [0, 1]$.

Practice questions.

1. *Numerical*. Find the mixed NE of Matching Pennies with payoffs scaled by 5. Does scaling change the probabilities? (No.)
2. *Equilibrium*. In the $n = 2$ crime game, verify $p^* = 1 - c/v$ from the general formula.
3. *Dominance*. Decide whether $M = (5, 5)$ is dominated by a mixture of $U = (10, 0)$ and $D = (0, 10)$.
4. *Graph*. Sketch BR_1, BR_2 for Battle of the Sexes and mark all three intersection points.

5. *MCQ*. A finite game with no pure NE: (a) has no NE; (b) is necessarily zero-sum; (c) still has a mixed NE; (d) is dominance-solvable. *[Ans: (c)]*

Week 4 — Games with Sequential Moves

The shift from tables to trees

Until now players moved *simultaneously*, so a table sufficed. Many real situations are *sequential*: one player moves, the other *sees* it, then responds. Order and observability matter. We draw such games as **trees** and solve them by reasoning *backward* from the end — “look forward, reason back.”

4.1 The Extensive Form (Game Trees)

Anatomy of a game tree

An **extensive-form** game is a tree with:

- a unique **initial node** (the root) where play begins;
- **decision nodes**, each owned by one player (or by “Nature” for chance moves);
- **branches** = *actions* available at a node;
- **terminal nodes** (leaves) carrying a **payoff** vector;
- **information sets** grouping nodes a player cannot tell apart.

Three structural rules: there is *only one* initial node; there are *no cycles*; and there is *exactly one path* from the root to each terminal node.

Two kinds of uncertainty

Chance / Nature’s move: exogenous randomness (a coin, the weather), modelled by a Nature node with fixed probabilities. **Strategic uncertainty**: not knowing what a *rational opponent* will choose. Information sets capture what a player does/does not observe.

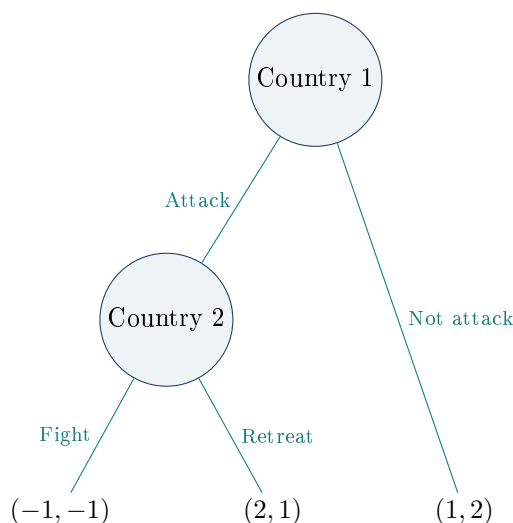
4.1.1 Perfect vs. imperfect information

Perfect information

A game has **perfect information** if every information set is a single node — i.e. whenever a player moves, she knows *exactly* where she is in the tree (all earlier moves are observed). If some information set contains ≥ 2 nodes (a player must move without knowing a prior choice), the game has **imperfect information**.

4.2 The War Game — a first tree

Country 1 chooses Attack (A) or Not (NA); if attacked, army 2 chooses Fight (F) or Retreat (R). Both prefer holding the island; a fight is worst for both.



Backward reasoning (next section): if attacked, army 2 compares Fight (-1) vs. Retreat (1) and *retreats*; anticipating this, Country 1 compares Attack $\rightarrow 2$ vs. Not $\rightarrow 1$ and *attacks*. Outcome $(2, 1)$.

4.3 Backward Induction (Rollback)

Backward induction / rollback equilibrium

In a finite perfect-information game, solve from the leaves upward: at each decision node, assume the owner picks the action leading to her best *eventual* payoff; replace the node by that payoff; repeat until the root. The resulting strategy profile is the **rollback (backward-induction) equilibrium**.

Equilibrium Insight

Backward induction is the disciplined form of “look forward, reason back.” In finite games of perfect information it yields a strategy profile that is a Nash equilibrium *and* more — it is *subgame perfect* (below), ruling out equilibria that rely on non-credible threats.

Common Mistake

Backward induction needs a *finite* horizon and well-defined ends. With infinite or ill-founded trees it can fail; the slides flag “not-finite games” as a caveat. Also: solve children *completely* before collapsing a parent — folding too early loses information.

4.4 Strategies in Extensive Form

Strategy in a tree

A **strategy** for player i specifies an action at *every* information set she owns — *even at nodes that will never be reached* in equilibrium. If a player owns information sets with $a_1, a_2, \dots$ actions respectively, she has $a_1 \times a_2 \times \dots$ pure strategies. This is why a player can have many more *strategies* than *actions*.

Successor / predecessor; subgame

Node y is a **successor** of x if x lies on the path from the root to y (x is then a *predecessor* of y). A **subgame** is the portion of the tree starting at a node x such that x and *all its successors* contain no information set that also includes nodes outside this portion. Equivalently: a subgame is a subtree whose root is *alone in its information set* and which is “self-contained.” The whole game is a (trivial) subgame of itself.

4.5 Subgame-Perfect Equilibrium and Credible Threats

Simple Intuition

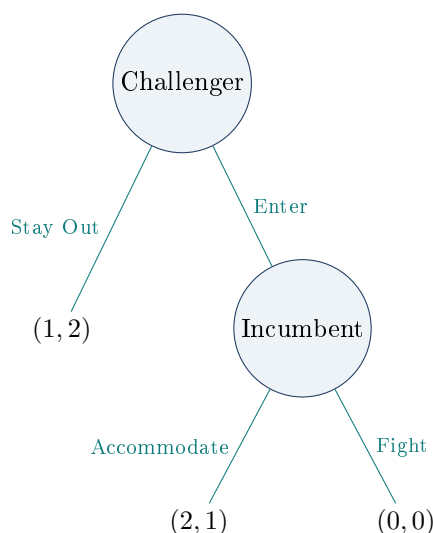
Some Nash equilibria of a tree are sustained by *threats that no rational player would carry out*. Subgame perfection throws those out: a prediction must be an equilibrium *in every subgame*, including ones never reached on the equilibrium path.

Subgame-perfect equilibrium (SPE)

A strategy profile is a **subgame-perfect equilibrium** if it induces a Nash equilibrium in *every* subgame (whether or not that subgame is reached in equilibrium). In finite perfect-information games, the SPE are exactly the backward-induction outcomes. **Every SPE is a Nash equilibrium; not every Nash equilibrium is subgame perfect.**

4.5.1 The Entry Game (credibility made concrete)

A Challenger chooses Stay Out or Enter; if it enters, the Incumbent chooses Accommodate or Fight.



Normal-form view (Challenger rows, Incumbent columns):

		Incumbent	
		Accommodate	Fight
Challenger	Stay Out	1, 2	1, 2
	Enter	2, 1	0, 0

Nash equilibria: (Enter, Accommodate) and (Stay Out, Fight). But “Fight” is *not credible*: if entry actually happens, the incumbent prefers Accommodate ($1 > 0$). Backward induction

keeps only

SPE = (Enter, Accommodate), outcome (2, 1).

The Nash equilibrium (Stay Out, Fight) survives only because of an *empty threat*; subgame perfection removes it.

4.5.2 The Ultimatum Game (bargaining by backward induction)

Simple Intuition

A Proposer is given a pie (say Rs. 100, or 1 unit) and offers a split: “I keep $1 - x$, you get x .” The Responder either *accepts* (the split stands) or *rejects* (both get 0). What does rationality predict?

Backward induction. At the Responder’s node, any offer $x > 0$ beats rejecting ($x > 0$), so the Responder accepts *every* strictly positive offer; she is indifferent only at $x = 0$. Anticipating this, the Proposer offers the *smallest acceptable amount* — in the limit $x \rightarrow 0$ — and keeps almost the entire pie.

SPE: Proposer offers (essentially) 0; Responder accepts.

Theory vs. the lab

The subgame-perfect prediction is a near-total split in the Proposer’s favour. In experiments, Responders routinely *reject* low offers (punishing unfairness at a cost to themselves) and Proposers offer 40–50%. This gap is the most famous evidence that real preferences include *fairness/reciprocity*, not just own-payoff maximisation — a standard exam talking-point and a limitation of pure backward induction with selfish payoffs.

4.5.3 When backward induction is delicate

“Not-finite” games

Backward induction needs a well-defined *end* to roll back from. In games with *infinite* horizons (or no last move) there is no final node to start at, so plain rollback does not directly apply; one uses stationarity or one-shot-deviation arguments instead. Always confirm the tree is finite before rolling back.

★ How to count things in a tree (the quiz loves this)

Terminal nodes = number of leaves = number of complete plays. **Information sets** = number of distinct “decision points” (singletons in perfect-info games). **Pure strategies of player i** = product of the number of actions over *all* information sets i owns. **Subgames** = number of nodes that are alone in their information set and start a self-contained subtree (the root counts; in a perfect-information line each internal node starts one).

4.6 First-Mover Advantage and the Value of Commitment

Simple Intuition

Moving first can be an *advantage* precisely because it is irreversible: by committing, you change the game the second mover faces. This is why a firm builds capacity before rivals, a country burns its bridges, or a negotiator makes a public “take-it-or-leave-it” stance. Commitment turns a would-be empty threat into a credible one.

Stackelberg (sequential Cournot). A leader picks quantity Q_1 first; the follower observes it and best-responds $Q_2 = \text{BR}_2(Q_1) = \frac{a-c}{2b} - \frac{Q_1}{2}$. The leader substitutes the follower’s reaction into its own profit and maximises:

$$\pi_1 = \left[a - c - b\left(Q_1 + \frac{a-c}{2b} - \frac{Q_1}{2}\right) \right] Q_1 = \frac{1}{2}(a-c)Q_1 - \frac{b}{2}Q_1^2 \Rightarrow Q_1^* = \frac{a-c}{2b}, \quad Q_2^* = \frac{a-c}{4b}.$$

The leader produces *more* than in simultaneous Cournot ($\frac{a-c}{3b}$) and earns *more*; the follower produces less. **Commitment to a large quantity is profitable because the follower rationally accommodates it** — the essence of first-mover advantage.

Equilibrium Insight

Contrast with simultaneous play: in Cournot neither firm can commit, so both land on the symmetric $\frac{a-c}{3b}$. Observability + irreversibility is what gives the leader power; if the follower *couldn’t* see Q_1 , the advantage vanishes. Commitment has value only when it is *visible* and *binding*.

4.7 The Centipede Game (a backward-induction paradox)

Simple Intuition

Two players alternate; at each turn a player can *Take* the larger share of a growing pot (ending the game) or *Pass* to let the pot grow. Mutual passing would make both far richer than immediate taking — yet backward induction predicts *Take at the very first move*.

Rolling back from the last node, whoever moves there Takes (a bird in hand); anticipating this, the previous mover Takes; ... unravelling all the way to the first node. The unique SPE is “Take immediately,” even though both would prefer to pass for several rounds.

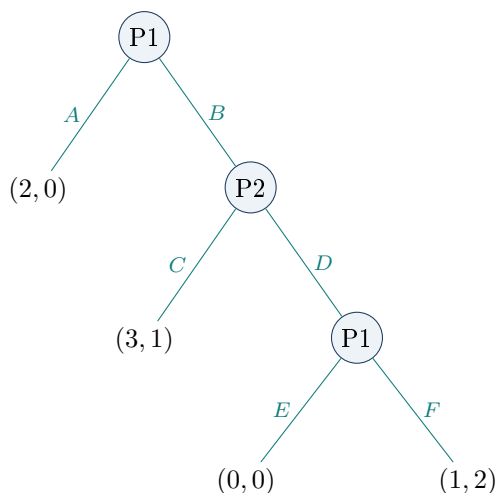
What it teaches

Centipede dramatises two things: (i) backward induction can yield a Pareto-poor outcome that clashes with intuition and with experiments (real players *do* pass for a while, sustained by trust/reputation); and (ii) the prediction hinges on *common knowledge of rationality at every node* — a single doubt about the opponent’s rationality can rationally justify passing. A standard “limitations of backward induction” talking point.

4.8 The Recurring Quiz Tree: A/B–C/D–E/F

Simple Intuition

This exact game appears in *every* Quiz 1 paper. Player 1 chooses *A* (game ends) or *B* (continue). After *B*, Player 2 chooses *C* (ends) or *D* (continue). After *D*, Player 1 chooses *E* or *F* (ends). Payoffs: $A \rightarrow (2, 0)$, $C \rightarrow (3, 1)$, $E \rightarrow (0, 0)$, $F \rightarrow (1, 2)$.



We will fully exploit this tree in Part II; here just note its anatomy: it is a game of **perfect information**, with **3 terminal nodes**, **3 information sets** (one per decision point), Player 1 owning *two* of them.

4.9 Bertrand Competition (price competition)

Simple Intuition

Cournot firms choose quantities; **Bertrand** firms choose *prices* simultaneously, and consumers buy from the cheaper one. With identical products this drives price to marginal cost (the “Bertrand paradox”); with differentiated demand it gives an interior Nash equilibrium found, again, by best responses.

Differentiated setup (the form the quiz uses). Two firms set prices $P_1, P_2 \geq 0$. Demand for firm i is $q_i = \alpha - P_i + P_j$ and production is costless, so profit is

$$\pi_i = (\alpha - P_i + P_j) P_i.$$

Maximising over P_i (treating P_j as given):

$$\frac{\partial \pi_i}{\partial P_i} = \alpha - 2P_i + P_j = 0 \implies \text{BR}_i(P_j) = \frac{\alpha + P_j}{2} = \frac{\alpha}{2} + \frac{P_j}{2}.$$

Symmetric equilibrium $P_1^* = P_2^* = P^*$: $P^* = \frac{\alpha}{2} + \frac{P^*}{2} \implies \frac{1}{2}P^* = \frac{\alpha}{2} \implies P^* = \alpha$.

★ The 2025 T3 Bertrand numbers — and the deliberate trap

There demand is $q_i = 10 - P_i + P_j$ (so $\alpha = 10$). The reaction function is $\text{BR}_i(P_j) = 5 + \frac{P_j}{2}$, and the symmetric NE solves $P = 5 + \frac{P}{2} \implies \frac{P}{2} = 5 \implies P^* = \mathbf{10}$ (*not 15!*). The exam asks for the *incorrect* statement and the keyed answer is exactly “*under the Nash equilibrium each player chooses $P = 15$* ” — because the true equilibrium price is 10. The game *is* symmetric, *does* have a (unique) symmetric NE; only the “ $P = 15$ ” claim is false. Full working in Part II.

Common Mistake

$P = 5 + \frac{P}{2}$ gives $\frac{P}{2} = 5$, hence $P = 10$. Writing $P = 15$ (e.g. by mis-solving $\frac{P}{2} = 5$ as $P = 15$) is the single most common error here — and it is precisely the distractor the paper plants.

End of Week 4

Feature	Simultaneous (normal form)	Sequential (extensive form)
Representation	Payoff matrix	Game tree
Timing	Move at once / in ignorance	Move in order; later see earlier
Information	Often imperfect by construction	Perfect if all past moves observed
Solution concept	Nash equilibrium	Subgame-perfect equilibrium
Tool	Underline best responses	Backward induction
Key issue	Coordination / beliefs	Credibility of threats

	Nash equilibrium	Subgame-perfect equilibrium
Definition	No profitable unilateral deviation on the equilibrium path	NE in <i>every</i> subgame (on <i>and</i> off path)
Allows non-credible threats?	Yes	No
How to find	Normal form + underline	Backward induction
Relationship	Every SPE is a NE	Not every NE is an SPE
Count (quiz tree)	3	1

Quick Revision — Week 4

Extensive form = tree (root, decision nodes, branches = actions, leaves = payoffs, information sets). **Perfect information** = every info set a singleton. **Strategy** = action at *every* info set (so #strategies = product of action counts). **Backward induction** solves perfect-info finite trees. **Subgame** = self-contained subtree with a singleton root. **SPE** = NE in every subgame = backward-induction outcome; rules out non-credible threats. **Bertrand** ($q_i = \alpha - P_i + P_j$): $BR_i = \frac{\alpha}{2} + \frac{P_j}{2}$, symmetric $P^* = \alpha$ (so $\alpha = 10 \Rightarrow P^* = 10$).

Counting cheat-sheet for trees

$$\#strategies_i = \prod_{\text{info sets } i \text{ owns}} (\#actions \text{ there}), \quad \#terminal \text{ nodes} = \#plays.$$

For the A/B–C/D–E/F tree: P1 owns 2 info sets ($\{A, B\}$ and $\{E, F\}$) $\Rightarrow 2 \times 2 = 4$ strategies; P2 owns 1 ($\{C, D\}$) $\Rightarrow 2$ strategies.

★ Algorithm sheet — solving a tree

(1) Draw/label the tree; identify each player's info sets. (2) Count leaves (terminal nodes). (3) From the deepest decision nodes, pick each owner's best action; collapse. (4) Continue to the root for the SPE path. (5) For Nash-but-not-SPE, also write the normal form and find all NE; the ones using non-credible off-path actions are the non-SPE.

Practice questions.

1. *Counting*. In the Entry Game, how many pure strategies does each player have? [*Challenger 2, Incumbent 2.*]
2. *Backward induction*. Re-solve the War Game if Fight gives army 2 a payoff of 3 instead of -1 . Does Country 1 still attack?
3. *SPE vs NE*. Explain why (Stay Out, Fight) is Nash but not subgame perfect.

4. *Bertrand*. With $q_i = 20 - P_i + P_j$, find BR_i and the symmetric NE price.
5. *Concept*. Give a tree with an information set of two nodes and state why it has imperfect information.

Part II

Complete PYQ Solutions

How Part II Works

Across the four Quiz-1 papers (2024 T2, 2024 T3, 2025 T2, 2025 T3) the same handful of *question families* recur, with numbers changed. Each section below takes one family and solves it in six steps:

1. **Interpretation** — restating the scenario as a precise game.
2. **Topic** — which Week's tool applies.
3. **Thought process** — how an expert decides what to do *before* computing.
4. **Full working** — every line of algebra.
5. **Traps** — the specific wrong answers the paper is engineered to catch.
6. **Verified answer** — the result, cross-checked against the official key.

Family 1 — The Auction / IESDS

Question

Appears in: multiple papers

A two-bidder, discrete-bid auction that secretly tests *dominance* (strict vs. weak, and *by a mixture*) and *iterated elimination*. The trap is to “obviously” delete bidding 0.

5.1 Interpretation

Bidder 1 values the object at 3; bidder 2 values it at 5 (values are common knowledge). Each *simultaneously* submits a bid from $\{0, 1, 2\}$. The **higher** bidder wins and pays *her own bid* (first-price); the loser pays nothing. **On a tie a fair coin decides the winner**, who then gets the object and pays his bid (the loser pays nothing). So on a tie each bidder wins with probability $\frac{1}{2}$ and the *expected* payoff is $\frac{1}{2}(v_i - b_i)$. Payoff = value – price if you win.

$$N = \{1, 2\}, \quad A_1 = A_2 = \{0, 1, 2\}, \quad u_i = \begin{cases} v_i - b_i, & b_i > b_j \\ \frac{1}{2}(v_i - b_i), & b_i = b_j \\ 0, & b_i < b_j. \end{cases}$$

5.2 Topic

Week 1 (dominance, strict vs. weak) and Week 2 (IESDS).

5.3 Thought process

The tie rule matters enormously. With a *coin toss*, a tie still delivers half the surplus, so a low bid is no longer “safe.” The examiner wants you to (a) see that bidder 2’s lowest bid is strictly dominated, while bidder 1 (the low-value bidder) has *no* strictly dominated pure strategy, and (b) run iterated elimination to a single surviving profile.

5.4 Full working

Build the payoff matrix with the coin-toss rule. Rows = bidder 1’s bid, columns = bidder 2’s bid; entry = (u_1, u_2) with $v_1 = 3$, $v_2 = 5$.

		Bidder 2		
		$b_2=0$	$b_2=1$	$b_2=2$
Bidder 1	$b_1=0$	1.5, 2.5	0, 4	0, 3
	$b_1=1$	2, 0	1, 2	0, 3
	$b_1=2$	1, 0	1, 0	0.5, 1.5

(E.g. $(0,0)$: tie, $u_1 = \frac{1}{2} \cdot 3 = 1.5$, $u_2 = \frac{1}{2} \cdot 5 = 2.5$. $(1,0)$: bidder 1 wins, pays 1, nets 2. $(1,1)$: tie, $u_1 = \frac{1}{2}(3-1) = 1$, $u_2 = \frac{1}{2}(5-1) = 2$. $(2,2)$: tie, $u_1 = \frac{1}{2}(3-2) = 0.5$, $u_2 = \frac{1}{2}(5-2) = 1.5$.)

Step 1 — does each player have a strictly dominated pure strategy?

Bidder 2 (read u_2 down each column):

$$b_2=0 : (2.5, 0, 0), \quad b_2=1 : (4, 2, 0), \quad b_2=2 : (3, 3, 1.5).$$

Compare $b_2=2$ against $b_2=0$: $3 > 2.5$, $3 > 0$, $1.5 > 0$ — $b_2=2$ **strictly dominates** $b_2=0$. So bidder 2 *does* have a strictly dominated strategy (bidding 0).

Bidder 1 (read u_1 across each row): $b_1=0 : (1.5, 0, 0)$, $b_1=1 : (2, 1, 0)$, $b_1=2 : (1, 1, 0.5)$. Checking all pairs, *no* row is strictly dominated by another single row (e.g. $b_1=0$ beats $b_1=2$ in column 0: $1.5 > 1$; $b_1=1$ beats $b_1=2$ in column 0 but loses in column 2). So **bidder 1 has no strictly dominated (pure) strategy**.

Answer to Q149

Player 1 is the player who does *not* have a strictly dominated strategy. (Player 2's bid of 0 is strictly dominated by bidding 2.)

Step 2 — iterated elimination of strictly dominated strategies

1. Delete bidder 2's $b_2=0$ (strictly dominated by $b_2=2$). Columns left: $b_2=1, 2$.
2. Now bidder 1's $b_1=0 = (0, 0)$ is strictly dominated by $b_1=2 = (1, 0.5)$ ($1 > 0$, $0.5 > 0$). Delete $b_1=0$. Rows left: $b_1=1, 2$.
3. Among $b_2=1, 2$ for bidder 2 over rows $b_1=1, 2$: $b_2=1 = (2, 0)$ vs $b_2=2 = (3, 1.5)$ gives $3 > 2$, $1.5 > 0$, so $b_2=2$ strictly dominates $b_2=1$. Delete $b_2=1$. Only $b_2=2$ remains.
4. Finally, against $b_2=2$, bidder 1 compares $b_1=1$ ($u_1 = 0$) with $b_1=2$ ($u_1 = 0.5$); $b_1=2$ wins. Delete $b_1=1$.

The unique survivor is $(b_1, b_2) = (2, 2)$.

Answer to Q150

$(2, 2)$ is the only strategy profile that survives IESDS. Both bidders bid their maximum, 2.

The tie-rule trap

The answer hinges on the *coin-toss* tie rule. Under it, a tie still pays half the surplus, which is what makes bidding 2 strictly beat bidding 0 for the high-value bidder 2 and drives the elimination all the way to $(2, 2)$. A student who assumes “ties pay nothing” gets no strict dominance and the wrong answer. **Read the tie sentence first.**

Common Mistake

Note bidder 1 has *no* strictly dominated pure strategy, yet IESDS *still* terminates at a single profile — because each *round* of elimination (driven initially by bidder 2) exposes a fresh strict domination for bidder 1. Eliminations alternate between the players; don't stop after checking only round one.

Family 2 — The Hawk–Dove Question (2024 T2)

Frequently Asked PYQ Concept

Identify which outcomes *cannot* be a pure Nash equilibrium, and find the equilibria of an anti-coordination game.

6.1 Interpretation & Topic

Two animals contest a resource of value $v = 10$. Fighting risks injury cost $k = 6$; displaying costs $d = 4$. Two Hawks split the value and share the fight cost; two Doves split the value minus display cost; Hawk vs. Dove $\Rightarrow$ Hawk takes all. **Topic:** Week 2 (classic 2×2 , Nash equilibrium); Week 3 (mixed NE).

6.2 Full working

With $v = 10, k = 6, d = 4$: $\frac{v}{2} - k = 5 - 6 = -1$, $\frac{v}{2} - d = 5 - 4 = 1$.

		Player 2	
		Hawk	Dove
Player 1	Hawk	$-1, -1$	10, 0
	Dove	0, 10	1, 1

Best-response check (underline method).

- If opponent plays Hawk: my Dove (0) beats my Hawk (-1) $\Rightarrow$ best response is **Dove**.
- If opponent plays Dove: my Hawk (10) beats my Dove (1) $\Rightarrow$ best response is **Hawk**.

So each player wants to do the *opposite* of the other. The doubly-best-response cells are the off-diagonal ones.

$$\text{Pure NE} = \{(\text{Hawk}, \text{Dove}), (\text{Dove}, \text{Hawk})\}.$$

Therefore the outcomes that **cannot** be pure NE are the diagonal ones: (Hawk, Hawk) and (Dove, Dove).

Mixed equilibrium (often the follow-up)

Let the opponent play Hawk with probability q . Indifference for the row player:

$$\underbrace{-q + 10(1 - q)}_{\mathbb{E}u(\text{Hawk})} = \underbrace{0 \cdot q + 1(1 - q)}_{\mathbb{E}u(\text{Dove})} \Rightarrow 10 - 11q = 1 - q \Rightarrow 9 = 10q \Rightarrow q^* = 0.9.$$

By symmetry $p^* = 0.9$: each plays **Hawk with probability 0.9** in the mixed NE.

Verified answer

Cannot be pure NE: (Hawk, Hawk) and (Dove, Dove). Pure NE are the two asymmetric profiles; the symmetric mixed NE is $p^* = q^* = 0.9$.

Quiz Trap

Hawk–Dove is *anti*-coordination: the pure equilibria are **off** the diagonal. Do not pattern-match to Stag Hunt / Battle of the Sexes, whose pure equilibria sit **on** the diagonal.

Family 3 — The Drug-Dealer Mixed-Strategy Question

The highest-value question in the paper

A constant-sum “pursuit” game (Police vs. Dealer) with *no* pure NE. You must (a) argue no pure NE exists and (b) compute the unique mixed NE. Appears with 100 trades (2024) and 500 trades (2025).

7.1 Interpretation & Topic

Police choose where to patrol (Street/Park); the Dealer chooses where to deal (Street/Park). Over a fixed number of trades the payoffs are how many the Police catch vs. the Dealer completes — they sum to a constant, so it is a **constant-sum** game. **Topic:** Week 3 (mixed strategies, indifference principle).

7.2 Version A — 100 trades (2024 T2)

Payoffs (Police, Dealer):

		Dealer	
		Street	Park
Police	Street	80, 20	0, 100
	Park	10, 90	60, 40

Step 1 — no pure NE

Best responses: if Dealer=Street, Police prefer Street ($80 > 10$); if Dealer=Park, Police prefer Park ($60 > 0$). If Police=Street, Dealer prefer Park ($100 > 20$); if Police=Park, Dealer prefer Street ($90 > 40$). The arrows chase each other in a cycle $\Rightarrow$ **no pure NE**.

Step 2 — mixed NE by indifference

Let Police play Street with prob. p , Dealer play Street with prob. q .

Police’s mix makes the Dealer indifferent. Dealer’s expected payoffs:

$$Eu_D(\text{Street}) = 20p + 90(1 - p) = 90 - 70p, \quad Eu_D(\text{Park}) = 100p + 40(1 - p) = 40 + 60p.$$

Set equal: $90 - 70p = 40 + 60p \Rightarrow 50 = 130p \Rightarrow p^* = \frac{5}{13}$ (Street), so Park = $\frac{8}{13}$.

Dealer's mix makes the Police indifferent. Police's expected payoffs:

$$\mathbb{E}_{UP}(\text{Street}) = 80q + 0(1 - q) = 80q, \quad \mathbb{E}_{UP}(\text{Park}) = 10q + 60(1 - q) = 60 - 50q.$$

Set equal: $80q = 60 - 50q \Rightarrow 130q = 60 \Rightarrow q^* = \frac{6}{13}$ (Street), so Park = $\frac{7}{13}$.

Verified answer — 100 trades

Unique (mixed) NE: Police (Street, Park) = $(\frac{5}{13}, \frac{8}{13})$, Dealer (Street, Park) = $(\frac{6}{13}, \frac{7}{13})$.
No pure NE exists.

7.3 Version B — 500 trades (2025 T2)

Payoffs (Police, Dealer):

		Dealer	
		Street	Park
Police	Street	400, 100	0, 500
	Park	100, 400	300, 200

Police's mix (Dealer indifferent):

$$100p + 400(1 - p) = 500p + 200(1 - p) \Rightarrow 400 - 300p = 200 + 300p \Rightarrow p^* = \frac{1}{3} \text{ (Street)}, \frac{2}{3} \text{ (Park)}.$$

Dealer's mix (Police indifferent):

$$400q + 0(1 - q) = 100q + 300(1 - q) \Rightarrow 400q = 300 - 200q \Rightarrow q^* = \frac{1}{2} \text{ (Street)}, \frac{1}{2} \text{ (Park)}.$$

Verified answer — 500 trades

Police = $(\frac{1}{3}, \frac{2}{3})$, Dealer = $(\frac{1}{2}, \frac{1}{2})$. No pure NE.

The cross-over trap — memorise this

Your probability is determined by the *opponent's* indifference. Police's p comes from setting the **Dealer's** two payoffs equal; Dealer's q comes from setting the **Police's** two payoffs equal. Pairing "Police payoffs" with "Police probability" is the most common way to lose this question.

Common Mistake

Always state *which* column/row each probability refers to (here "Street"). Reporting $\frac{8}{13}$ when the key wants the *Street* probability $\frac{5}{13}$ is a needless error. And check the two numbers sum to 1.

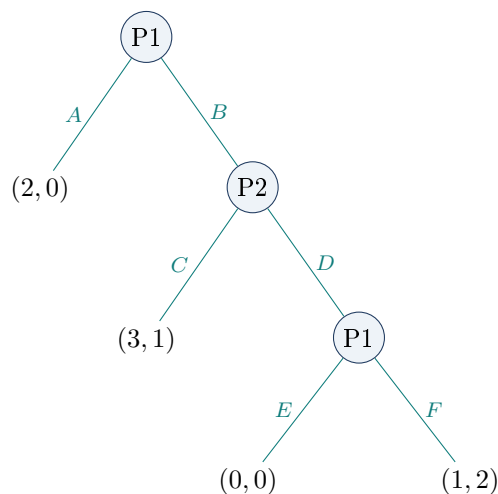
Family 4 — The Sequential A/B–C/D–E/F Tree

In every paper

Count the tree's features (terminal nodes, information sets, each player's number of strategies), then find the pure-strategy Nash equilibria and the subgame-perfect equilibrium.

8.1 Interpretation & Topic

Player 1 chooses A (ends) or B . After B , Player 2 chooses C (ends) or D . After D , Player 1 chooses E or F (ends). Payoffs $A \rightarrow (2, 0)$, $C \rightarrow (3, 1)$, $E \rightarrow (0, 0)$, $F \rightarrow (1, 2)$. **Perfect information.** **Topic:** Week 4 (extensive form, backward induction, SPE).



8.2 Counting (the MCQ part)

- **Terminal nodes (leaves):** the plays end at A , at C , at E , and at F — so there are **4** terminal nodes. (*Keyed answer: 4.*)
- **Information sets:** one at the root (P1), one after B (P2), one after D (P1) = **3**, all singletons (perfect information).
- **Strategies:** P1 owns *two* information sets ($\{A, B\}$ and $\{E, F\}$) $\Rightarrow 2 \times 2 = 4$ strategies: AE, AF, BE, BF . P2 owns *one* ($\{C, D\}$) $\Rightarrow 2$ strategies: C, D .

- **Subgames:** in a perfect-information tree, every decision node initiates a subgame. The three decision nodes (root, P2's node, P1's second node) each start one self-contained subtree, so the game has **3 subgames** (the whole game counts as a subgame of itself; the other two are the proper subgames after B and after D). (*Keyed answer: 3.*)

★ Exam Tip

“Terminal nodes” = leaves you can actually end at (here $A, C, E, F \Rightarrow 4$). “Strategies” = product of action-counts over a player's information sets, *counting nodes that are never reached on the equilibrium path* — that off-path counting is why P1 has 4 (not 2) strategies. “Subgames” = number of decision nodes alone in their information set (here 3).

8.3 Backward induction (the SPE)

1. **Last node (P1, after D):** $E \rightarrow 0$ vs $F \rightarrow 1$. P1 picks F ; value $(1, 2)$.
2. **P2's node:** $C \rightarrow (3, 1)$ vs $D \rightarrow (1, 2)$. P2 compares *own* payoff 1 vs 2 and picks D ; value $(1, 2)$.
3. **Root (P1):** $A \rightarrow (2, 0)$ vs $B \rightarrow (1, 2)$. P1 compares 2 vs 1 and picks A ; value $(2, 0)$.

SPE strategies: $P1 = (A, F)$, $P2 = D$; outcome $(2, 0)$.

8.4 All pure Nash equilibria (the harder part)

Write the normal form. Rows = P1's four strategies, columns = P2's two; entries (u_1, u_2) :

		Player 2	
		C	D
Player 1	AE	2, 0	2, 0
	AF	2, 0	2, 0
	BE	3, 1	0, 0
	BF	3, 1	1, 2

(If P1's plan starts with A the game ends at the root, so both columns give $(2, 0)$.) Underline best responses:

- Column C : P1's best payoff is 3 (BE, BF). Column D : P1's best is 2 (AE, AF).
- Row BE : P2 prefers C ($1 > 0$). Row BF : P2 prefers D ($2 > 1$). Rows AE, AF : P2 is indifferent ($0 = 0$), so D is (weakly) a best response.

Doubly-best cells: (BE, C) , (AE, D) , (AF, D) .

Three pure NE: $(BE, C) \rightarrow (3, 1)$, $(AE, D) \rightarrow (2, 0)$, $(AF, D) \rightarrow (2, 0)$.

Which are subgame perfect?

Only (AF, D) survives: it prescribes F at the last node (credible, $1 > 0$) and D for P2 (credible, $2 > 1$).

- (AE, D) is *not* SPE: at the last node E is not optimal ($0 < 1$) — a non-credible plan that is never tested because play ends at A .
- (BE, C) is *not* SPE: it relies on P1 threatening E , which fails subgame perfection at the last node; given a credible F , P2 would switch to D .

Verified answer

Terminal nodes **4** (ends A, C, E, F); information sets **3**; subgames **3**; strategies **4** (P1) and **2** (P2). **Pure NE: 3** — $(BE, C), (AE, D), (AF, D)$. **Unique SPE** = $(A, F; D)$ with outcome $(2, 0)$.

Quiz Trap

The number of *pure NE* (3) exceeds the number of *SPE* (1). If a question asks “how many Nash equilibria,” it wants all three; if it asks for the SPE / backward-induction outcome, it wants only $(2, 0)$. Mark which is being asked.

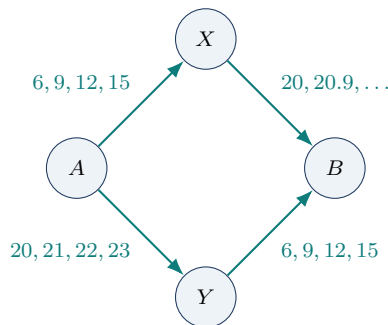
Family 5 — The Congestion / Braess Routing Question

Frequently Asked PYQ Concept

“ N drivers choose a route” — find the equilibrium travel time, then add a new link and show everyone is worse off. Tests the Nash logic of *equalising route times*.

9.1 Interpretation & Topic

Four people drive $A \rightarrow B$ at once. Two routes: $A-X-B$ and $A-Y-B$. Edges $A-X$ and $Y-B$ are *short & narrow*: one car 6 min, each *additional* car adds 3 min/car (so m cars $\Rightarrow 6+3(m-1)$; the figure lists 6, 9, 12, 15 for $m = 1, 2, 3, 4$). Edges $A-Y$ and $X-B$ are *long & wide*: $A-Y$ is $20 + 1(m-1)$ (i.e. 20, 21, 22, 23), $X-B$ is $20 + 0.9(m-1)$ (i.e. 20, 20.9, 21.8, 22.7). **Topic:** Week 2 (Nash in a congestion game). The master technique: *at a pure-strategy equilibrium every route actually used has (essentially) the same travel time*.



9.2 Base game (Q307 / Q95 / Q279)

Try **two drivers per route**. On $A-X-B$: $A-X$ has 2 cars $\Rightarrow 6 + 3 = 9$; $X-B$ has 2 cars $\Rightarrow 20 + 0.9 = 20.9$; total $T_{AXB} = 9 + 20.9 = \mathbf{29.9}$. On $A-Y-B$: $A-Y$ has 2 cars $\Rightarrow 20 + 1 = 21$; $Y-B$ has 2 cars $\Rightarrow 6 + 3 = 9$; total $T_{AYB} = 21 + 9 = \mathbf{30}$. The two times (29.9 and 30) are essentially equal, and no driver can gain by switching: moving onto a route would make it carry 3 cars (e.g. $A-X$ becomes $6 + 6 = 12$, $X-B$ becomes $20 + 1.8 = 21.8$, total $33.8 > 30$). So “two per route” is the Nash equilibrium.

Verified answer — base case

In Nash equilibrium **two people take each route**, and each person's travel time is **either 29.9 or 30 minutes**. (Keyed: the combined statement is correct. Note the 2025 T2 paper's "28 minutes" option is *wrong* — the times are 29.9/30, so there only "two per route" is true.)

9.3 Add the road $X-Y$ (Q308 / Q96 / Q280)

A new short-wide link $X-Y$ costs $7 + 1(m-1)$ (7, 8, 9, 10). Four routes now exist; the tempting one is $A-X-Y-B$, using the two *cheap* congestible edges $A-X$ and $Y-B$. The Nash equilibrium is **1 driver on $A-X-B$, 2 on $A-X-Y-B$, 1 on $A-Y-B$** . Edge loads and per-car times:

$$\underbrace{A-X}_{3 \text{ cars}} = 12, \quad \underbrace{X-B}_{1 \text{ car}} = 20, \quad \underbrace{X-Y}_{2 \text{ cars}} = 8, \quad \underbrace{Y-B}_{3 \text{ cars}} = 12, \quad \underbrace{A-Y}_{1 \text{ car}} = 20.$$

Each used route then costs

$$T_{AXB} = 12 + 20 = 32, \quad T_{AXYB} = 12 + 8 + 12 = 32, \quad T_{AYB} = 20 + 12 = 32.$$

All three equal 32, and any deviation raises a cheap edge to 4 cars ($6 + 9 = 15$) and gives $\geq 36 > 32$ — so (1, 2, 1) is the equilibrium. **Travel time rises from ~ 30 to 32 minutes although a new road was added — Braess's paradox.**

Verified answer — after the new road

NE: one person $A-X-B$, two $A-X-Y-B$, one $A-Y-B$; every person's travel time is **32 minutes**, which is *higher* than before the road (so "travel time increases" is the true statement keyed in 2025 T2/T3).

Potential Issue Detected

The 2024 T3 option pairs the (correct) configuration with "each person's travel time is 36 minutes." By the edge costs given (and the figure's labels), the common equilibrium time is **32 minutes**, not 36 — and indeed the paper's key marks the "36 minutes" choice as *incorrect*, leaving the configuration-only statement as the right answer. Likewise, the 2025 T3 option that states the split as "*two* take $A-X-B$, one takes $A-X-Y-B$, one takes $A-Y-B$ " is *not* an equilibrium (its route times 32.9/28/29 are unequal); the only universally true 2025 T3 statement is "every person's travel time increases." *Please verify the intended travel-time figure with the course instructor.*

Quiz Trap

At equilibrium it is *travel times* that equalise across *used* routes, not the number of cars. Compute each edge's per-car time from " $6 + 3(m-1)$ "-type rules using the *total* cars on that edge (summed over every route that uses it), then add edges along each route.

Family 6 — The Bertrand Price-Competition Question (2025 T3)

Frequently Asked PYQ Concept

Derive the best-response (reaction) function for a differentiated-Bertrand duopoly and identify the symmetric Nash equilibrium price; then judge true/false statements about it.

10.1 Interpretation & Topic

Two firms set prices P_1, P_2 simultaneously; firm i faces demand $q_i = 10 - P_i + P_j$ and has zero cost. **Topic:** Week 4 / Week 2 (Nash equilibrium with continuous actions; best response by calculus).

10.2 Full working

Profit $\pi_i = P_i q_i = P_i(10 - P_i + P_j) = 10P_i - P_i^2 + P_i P_j$. Maximise over P_i (treat P_j fixed):

$$\frac{\partial \pi_i}{\partial P_i} = 10 - 2P_i + P_j = 0 \implies \boxed{\text{BR}_i(P_j) = 5 + \frac{1}{2}P_j}.$$

Symmetric equilibrium $P_1^* = P_2^* = P^*$:

$$P^* = 5 + \frac{1}{2}P^* \implies \frac{1}{2}P^* = 5 \implies \boxed{P^* = 10}.$$

Each firm's equilibrium quantity $q_i^* = 10 - 10 + 10 = 10$; profit $\pi_i^* = 10 \times 10 = 100$.

Verified answer

Q93: the reaction function is $\text{BR}_i(P_j) = 5 + \frac{1}{2}P_j$ (keyed correct).
Q94 (“choose the incorrect statement”): the keyed answer is “Under the Nash equilibrium each player chooses $P = 15$ ” — this statement is *false*. The true equilibrium price is $P^* = 10$. The other three statements are all true: the game *is* symmetric, it *has* a symmetric Nash equilibrium, and that equilibrium *is* unique.

Why $P = 15$ is wrong — and why it is offered

Solving $P = 5 + \frac{P}{2}$ correctly: subtract $\frac{P}{2}$ to get $\frac{P}{2} = 5$, so $P = 10$. A student who instead computes “ $P = 5 + \frac{P}{2} \Rightarrow$ multiply...” carelessly lands on 15 — which is exactly the distractor the paper marks as the incorrect statement. Also differentiate $\pi_i = P_i(10 - P_i + P_j)$ carefully: the middle term gives $-2P_i$ (not $-P_i$); dropping the factor of 2 gives the wrong $BR_i = 10 + P_j$.

Family 7 — The 2×2 Mixed-Equilibrium Numeric (2025 T3)

Frequently Asked PYQ Concept

A 2×2 game with no pure NE; find the unique mixed equilibrium and then compute a numeric expression of the probabilities (here $\frac{1}{p-q}$).

11.1 Interpretation & Topic

Player 1 chooses Top/Bottom; Player 2 chooses Left/Right. Payoffs (u_1, u_2) : (Top, Left) = (8, 2), (Top, Right) = (1, 6), (Bottom, Left) = (3, 5), (Bottom, Right) = (4, 1). **Topic:** Week 3 (mixed NE, indifference).

		Player 2	
		Left	Right
Player 1	Top	8, 2	1, 6
	Bottom	3, 5	4, 1

11.2 Full working

No pure NE (best responses cycle): vs. Left, P1 wants Top ($8 > 3$); vs. Top, P2 wants Right ($6 > 2$); vs. Right, P1 wants Bottom ($4 > 1$); vs. Bottom, P2 wants Left ($5 > 1$). The cycle confirms a *unique mixed* NE.

Let $p = \Pr[\text{P1 plays Top}]$, $q = \Pr[\text{P2 plays Left}]$.

P2 indifferent (this pins down p):

$$\mathbb{E}u_2(\text{Left}) = 2p + 5(1 - p) = 5 - 3p, \quad \mathbb{E}u_2(\text{Right}) = 6p + 1(1 - p) = 1 + 5p.$$

$$5 - 3p = 1 + 5p \Rightarrow 4 = 8p \Rightarrow \boxed{p = \frac{1}{2}}.$$

P1 indifferent (this pins down q):

$$\mathbb{E}u_1(\text{Top}) = 8q + 1(1 - q) = 1 + 7q, \quad \mathbb{E}u_1(\text{Bottom}) = 3q + 4(1 - q) = 4 - q.$$

$$1 + 7q = 4 - q \Rightarrow 8q = 3 \Rightarrow \boxed{q = \frac{3}{8}}.$$

The requested number:

$$p - q = \frac{1}{2} - \frac{3}{8} = \frac{4-3}{8} = \frac{1}{8} \Rightarrow \frac{1}{p - q} = \boxed{8}.$$

Verified answer

Unique mixed NE: $p = \Pr[\text{Top}] = \frac{1}{2}$, $q = \Pr[\text{Left}] = \frac{3}{8}$. Hence $\frac{1}{p - q} = 8$ (the keyed short-answer value).

Quiz Trap

Keep the cross-over straight: p (P1's probability) comes from **P2's** indifference; q (P2's probability) comes from **P1's** indifference. Also confirm a pure NE truly doesn't exist before mixing — if one did, the “mixed” answer would be wrong.

Family 8 — Numeric chapterFamily 8 — Stray Numeric Items (Annuity & Returns to Scale) Microeconomics Items (Annuity, Production Theory)

Frequently Asked PYQ Concept

The papers occasionally include a finance/annuity computation (2024 T3) and a block of microeconomics concept items — Cobb–Douglas returns to scale and the average-product/marginal-product relationship (2025 T2). They are quick marks if you know the formulas.

12.1 Annuity — “when do you become a millionaire?” (2024 T3)

Deposit Rs. 25,000 at the end of each year; interest 8% compounded annually. The future value of an n -year ordinary annuity is

$$FV = P \cdot \frac{(1+r)^n - 1}{r} = 25000 \cdot \frac{(1.08)^n - 1}{0.08}.$$

Set $FV \geq 1,000,000$:

$$\frac{(1.08)^n - 1}{0.08} \geq 40 \Rightarrow (1.08)^n \geq 4.2 \Rightarrow n \geq \frac{\ln 4.2}{\ln 1.08} = \frac{1.4351}{0.07696} \approx 18.65.$$

So you cross a million during year 19.

12.2 Production theory (2025 T2): Cobb–Douglas & AP–MP

Q274. For $F(K, L) = K^p L^q$, scaling inputs by t gives $t^{p+q}F$. So it shows *constant* returns to scale iff $p + q = 1$; it can show *decreasing* returns ($p + q < 1$), and its marginal products are *not* constant. Correct option: “constant returns to scale if $p + q = 1$.”

Q275–277 (AP vs MP). Using $\frac{d}{dL}(Q/L) = \frac{MP_L - AP_L}{L}$ (derived in Part III §13.5):

- “AP decreasing $\Rightarrow$ MP $>$ AP”: **FALSE** (decreasing AP $\Rightarrow$ MP $<$ AP).
- “AP increasing $\Rightarrow$ MP $<$ AP”: **FALSE** (increasing AP $\Rightarrow$ MP $>$ AP).
- “AP at maximum $\Rightarrow$ MP = AP”: **TRUE**.

12.3 Cobb–Douglas returns to scale (summary)

For $Q = AL^aK^b$, scaling both inputs by t gives $A(tL)^a(tK)^b = t^{a+b}Q$. Hence returns to scale are **increasing** if $a + b > 1$, **constant** if $a + b = 1$, **decreasing** if $a + b < 1$. (Just add the exponents.)

Verified answers

Annuity: **19 years**. Cobb–Douglas RTS: constant iff $p + q = 1$. AP/MP: **FALSE, FALSE, TRUE** for Q275–277.

★ Exam Tip

For annuities, decide *ordinary* (deposits at year-end, formula above) vs. *annuity-due* (deposits at year-start, multiply FV by $(1+r)$). The paper’s “at the end of each year” means ordinary. For AP/MP, remember the running-average analogy: a marginal value above the average pulls the average up.

Part III

Concepts Found Only in the PYQs

Gap-Filling Concepts

Why this Part exists

Some ideas are *tested* by Quiz 1 but barely *taught* by the slides. If you study only the lectures you will meet them for the first time in the exam. This Part collects them, each with the minimum theory needed to answer the corresponding question cold.

13.1 Constant-Sum Structure and Why It Forces Mixing

Constant-sum game

A two-player game is **constant-sum** if $u_1(a) + u_2(a) = k$ for the same constant k in every cell (zero-sum is the case $k = 0$). The drug-dealer game ($k = 100$ or 500) is constant-sum.

Equilibrium Insight

In a strictly competitive constant-sum game with no pure NE, the players' interests are exactly opposed, so any predictable pure choice is exploitable — pushing the unique equilibrium into *mixed* strategies. Recognising “the two payoffs always add to the same total” is your cue that (i) the game is pure conflict and (ii) you should look for a mixed NE via indifference. The equilibrium mix also has a *security* (minimax) interpretation: each player guarantees her value regardless of the opponent.

13.2 Domination by a Mixed Strategy (the auction's hidden idea)

Definition

A pure action b of player i is **strictly dominated by a mixed strategy** σ_i if

$$u_i(\sigma_i, a_{-i}) > u_i(b, a_{-i}) \quad \text{for every } a_{-i},$$

where $u_i(\sigma_i, a_{-i}) = \sum_x \sigma_i(x) u_i(x, a_{-i})$ is the expected payoff of the mixture.

Procedure to test it (two-mixing-actions case). To check if $b = (b^L, b^R)$ (payoffs against the opponent's two actions) is dominated by mixing $x = (x^L, x^R)$ and $y = (y^L, y^R)$ with weight α on x :

$$\text{need } \alpha x^L + (1 - \alpha)y^L > b^L \quad \text{and} \quad \alpha x^R + (1 - \alpha)y^R > b^R \quad \text{for some } \alpha \in [0, 1].$$

If the two inequalities give a non-empty range for α , b is dominated; if they conflict (e.g. “ $\alpha > 0.6$ and $\alpha < 0.4$ ”), it is not. (*This is the exact test used in Week 3’s worked example and underlies the auction question’s resolution.*)

★ Exam Tip

The crucial subtlety: a strategy that is *never strictly beaten by any single pure strategy* can still be strictly beaten by a *mixture* — and only then may IESDS remove it. Conversely, if even the best mixture ties in some column, the strategy is *not* strictly dominated and survives.

13.3 Congestion-Game Arithmetic (Braess in numbers)

Route-time bookkeeping

On a route, total time = $\sum_{\text{edges}} (\text{flat cost} + (\text{per-car cost}) \times (\text{cars on that edge}))$. At a pure NE every *used* route shows the *same* total time; an *unused* route’s time is $\geq$ that common value.

Solving recipe. (1) Let n take route 1, rest take route 2. (2) Write $T_1(n)$ and $T_2(n)$. (3) Solve $T_1(n) = T_2(n)$ for n (use the nearest feasible integer split, then verify no driver gains by switching). (4) After adding a link, find which edges *all* traffic now wants and recompute. The diagnostic of Braess: the post-link common time *exceeds* the pre-link time.

13.4 Time Value of Money: Annuities and Compounding

The four formulas you may need

Future value (lump sum): $FV = PV(1 + r)^n$.

Present value (lump sum): $PV = FV/(1 + r)^n$.

FV of ordinary annuity: $FV = P \frac{(1 + r)^n - 1}{r}$.

PV of ordinary annuity: $PV = P \frac{1 - (1 + r)^{-n}}{r}$.

Annuity-due (payments at the *start* of each period): multiply the ordinary-annuity value by $(1 + r)$.

Solving for n . Isolate $(1 + r)^n$, take logs: $n = \frac{\ln(\text{target})}{\ln(1 + r)}$, then round *up* to the next whole period (you reach the goal only after the period completes).

13.5 Production Theory: Returns to Scale and the AP–MP Relationship

Returns to scale

For a production function $Q = f(L, K)$, scaling all inputs by $t > 1$: *increasing* returns if $f(tL, tK) > t f(L, K)$; *constant* if $=$; *decreasing* if $<$. For Cobb–Douglas $Q = AL^a K^b$ (the paper writes $F(K, L) = K^p L^q$): $f(tL, tK) = t^{a+b}Q$, so compare the exponent sum with 1.

Thus it shows *constant* returns iff $p + q = 1$, and it can certainly exhibit *decreasing* returns (when $p + q < 1$) — so “never exhibits decreasing returns” is false, and Cobb–Douglas does *not* have a constant marginal product of either input.

Average product vs. marginal product of labour

Let $AP_L = Q/L$ and $MP_L = \partial Q/\partial L$. Differentiating AP_L :

$$\frac{d}{dL} \left(\frac{Q}{L} \right) = \frac{MP_L - AP_L}{L}.$$

Hence the universal relationship: AP_L rising $\iff MP_L > AP_L$; AP_L falling $\iff MP_L < AP_L$; AP_L at its maximum $\iff MP_L = AP_L$ (the MP curve cuts the AP curve at AP’s peak).

★ The three true/false items decoded (2025 T2)

“When AP is *decreasing*, MP is *greater* than AP” $\Rightarrow$ **FALSE** (decreasing AP means $MP < AP$). “When AP is *increasing*, MP is *less* than AP” $\Rightarrow$ **FALSE** (increasing AP means $MP > AP$). “When AP is at a maximum, MP equals AP” $\Rightarrow$ **TRUE**. Memory aid: “a new value above the average pulls the average up” — MP relates to AP exactly as a new test score relates to your running grade average.

13.6 Reading Tie-Breaking and “Who Wins” Rules

The single most answer-flipping detail

Auction and contest questions hinge on the *tie rule*: “object not awarded” vs. “split the surplus” vs. “lower-index bidder wins” give *different* domination structures. Read this sentence in the question *before* building the matrix; it determines whether bidding 0 is strictly dominated or merely weakly dominated.

13.7 Equivalences You Can Quote

Three theorem-level facts the quiz rewards

1. **Never-best-response** $\iff$ strictly dominated by a mixed strategy (so such actions get probability 0 in every NE).
2. **IESDS** is order-independent and preserves all Nash equilibria; **IEWDS** is order-dependent and can delete a NE.
3. **Every SPE is a NE**; backward induction = SPE in finite perfect-information games; the count of NE can exceed the count of SPE.

Part IV

Quick Revision Sheets

Last-Minute Revision

14.1 One-Page Summary (Weeks 1–4)

The whole syllabus on one screen

Game = $(N, (A_i), \text{info/rules}, (u_i))$; describe any scenario with **PAPI-R**. **Classify**: static/dynamic, complete/incomplete, cooperative/non-cooperative (Quiz 1 = non-cooperative, complete info, static & dynamic).

Solution-concept ladder (strong $\rightarrow$ weak): dominant-strategy equilibrium $\Rightarrow$ survives IESDS $\Rightarrow$ Nash equilibrium.

Pure NE: underline each player's best payoff in every row/column; doubly-underlined cells are NE.

Mixed NE: your randomisation makes the *opponent* indifferent; equate the opponent's two expected payoffs (cross-over). Finite games always have one (Nash).

Sequential games: draw the tree; *backward induction* gives the *SPE*; SPE kills non-credible threats. #strategies = product of action-counts over a player's info sets.

14.2 Equilibrium Cheat-Sheet

Concept	Definition (one line)	How to find it
Dominant strategy	Best <i>for all</i> opponent profiles	Compare a row/column against <i>every</i> alternative
IESDS survivor	Not removed by deleting strictly dominated actions	Iterate strict-dominance deletion (also test mixtures)
Pure Nash	Mutual best response; no profitable unilateral deviation	Underline best responses; double underline = NE
Mixed Nash	Players randomise; supports tie in payoff	Indifference: equate opponent's expected payoffs
SPE	NE in every subgame	Backward induction on the tree

14.3 Key Formula Card

Everything numeric you might need

Best response: $BR_i(a_{-i}) = \arg \max_{a_i} u_i(a_i, a_{-i})$.

Cournot: $BR_i(Q_j) = \frac{a-c_i}{2b} - \frac{Q_j}{2}$, $Q_i^* = \frac{a-c}{3b}$ (symmetric).

Bertrand (diff.): $BR_i(P_j) = \frac{\alpha}{2} + \frac{P_j}{2}$, $P^* = \alpha$ (here $\alpha = 10 \Rightarrow P^* = 10$).

Hotelling: $x_A^* = x_B^* = \frac{1}{2}$.

Bystander: $p^* = 1 - (c/v)^{1/(n-1)}$, $\Pr[\text{no call}] = (c/v)^{n/(n-1)}$.

Annuity (FV): $FV = P \frac{(1+r)^n - 1}{r}$, $n = \frac{\ln(\text{target})}{\ln(1+r)}$ (round up).

Returns to scale: $Q = AL^a K^b$: $a + b \gtrless 1 \Rightarrow \text{incr./const./decr.}$

14.4 Algorithm Cheat-Sheet

★ Pick the matching recipe under time pressure

Matrix, find pure NE: underline method.

Matrix, no pure NE / asked to mix: p, q on first actions $\rightarrow$ equate *Player 1's* two payoffs for q , *Player 2's* two payoffs for $p \rightarrow$ check $\in [0, 1]$.

Dominance / IESDS: delete strictly dominated (try mixtures); stop when none remain.

Tree: count info sets & leaves $\rightarrow$ backward-induct for SPE $\rightarrow$ (if asked for all NE) write normal form and underline.

Cournot/Bertrand: write profit $\rightarrow$ FOC $\rightarrow$ reaction function $\rightarrow$ solve simultaneously.

Routing: equate *used*-route times $\rightarrow$ solve split $\rightarrow$ recompute after new link.

14.5 Common Exam Traps (the “do-not-lose-these” list)

Top ten traps across the four papers

1. Calling a *weakly* dominated action “strictly dominated” (auction: bidding 0).
2. Forgetting that a strategy can be dominated only by a *mixture*.
3. Pairing your own probability with your own indifference (mixed NE cross-over).
4. Treating Hawk–Dove like a coordination game (pure NE are *off*-diagonal).
5. Reporting the wrong coordinate of a mixed strategy (Street vs. Park) or one that doesn't sum to 1.
6. Confusing “number of Nash equilibria” with “number of SPE” in the tree question.
7. Counting a player's *actions* instead of *strategies* (forgetting off-path info sets).
8. In Cournot/Bertrand, dropping the factor of 2 in the first-order condition.
9. Assuming a 50–50 split in an *asymmetric* routing game (equate times, not cars).
10. Ignoring the tie-breaking rule in auctions/contests — it can flip the answer.

14.6 Five-Minute Pre-Exam Notes

Read this on the way in

- Mixed-NE cross-over: *my* prob. $\leftarrow$ *your* indifference.
- Underline method finds *all* pure NE fast.
- Backward induction = SPE; $\text{SPE} \subseteq \text{NE}$.
- Anti-coordination (Hawk–Dove): equilibria off-diagonal; mixed $p^* = 0.9$ for $v=10, k=6, d=4$.
- Drug-dealer: $\frac{5}{13}, \frac{6}{13}$ (100 trades); $\frac{1}{3}, \frac{1}{2}$ (500 trades).
- Bertrand $q_i = 10 - P_i + P_j$: $\text{BR} = 5 + \frac{P_j}{2}$, $P^* = 10$ (the “ $P = 15$ ” option is the planted wrong answer).
- Tree A/B–C/D–E/F: SPE outcome (2,0); 3 pure NE, 1 SPE.
- Annuity Rs. 25,000 at 8%: millionaire in 19 years.

Part V

Exam Strategy Guide

How to Attack Quiz 1

15.1 The 60-Second Triage

Simple Intuition

Every Quiz-1 question is one of a few types. Spend the first minute *classifying*, not solving. The class determines the recipe; the recipe determines the marks.

If you see...	...do this
A payoff matrix, “equilibrium”	Underline best responses; report doubly-underlined cells.
“randomise”, no pure NE, constant-sum	Indifference method (cross-over); check probabilities sum to 1.
“dominated”, “survives”, auction bids	Strict-dominance test <i>including mixtures</i> ; watch weak vs. strict.
A tree / “sequential” / “credible”	Backward induction for SPE; normal form if “how many NE”.
“quantity”/“price”, two firms	FOC on profit → reaction function → solve.
“route”/“drivers”/“travel time”	Equate used-route times; recompute after new link.
“deposit”/“interest”/“value”	Time-value formula; solve for n with logs.

15.2 How to Approach MCQs

★ Exam Tip

Eliminate first: cross out options that violate a theorem (e.g. “a finite game with no NE” is impossible by Nash). For “which statement is false,” verify each against a computed fact rather than intuition. For counting questions (terminal nodes, strategies), recount on the drawn figure — never from memory.

15.3 How to Analyse Game Trees

1. Label every node with its owner and every branch with its action.
2. Count leaves (terminal nodes) and each player’s information sets.

3. Strategies = product of action-counts over a player's info sets (include off-path sets).
4. Backward-induct from the deepest nodes; record the chosen action and the value at each.
5. For “all Nash equilibria,” build the normal form and underline; remember some NE are not SPE.

15.4 How to Find Nash Equilibria Quickly

★ Exam Tip

Pure: the underline method beats checking every cell. Two-player $2 \times n$: eliminate strictly dominated columns first to shrink the matrix. Always sanity-check a candidate NE by asking “can either player gain by switching?” — if no, it's a NE.

15.5 How to Solve Mixed-Strategy Problems

1. Confirm there is no pure NE (else you may not need to mix).
2. Assign p to the row player's first action, q to the column player's first.
3. Write each player's two expected payoffs as linear functions of the *opponent's* probability.
4. Equate *Player 1's* two payoffs $\rightarrow$ solve for q ; equate *Player 2's* $\rightarrow$ solve for p .
5. Verify $p, q \in [0, 1]$ and that supports tie. Report the requested coordinate/expression.

15.6 Time Management

Remember This

Budget by marks, not by order. Bank the quick wins first (underline-method NE, Bertrand/Cournot FOC, annuity) in the opening third. Reserve the back third for the two “heavy” items — the drug-dealer mixed NE and the tree's full NE-vs-SPE analysis — which carry the most marks and the most partial credit. Never leave a true/false blank: a verified fact usually settles it in seconds.

15.7 Common Mistakes to Avoid (condensed)

Common Mistake

Weak vs. strict dominance; the mixed-NE cross-over; off-diagonal Hawk–Dove equilibria; actions vs. strategies in trees; the factor of 2 in duopoly FOCs; assuming equal splits in asymmetric routing; ignoring tie-breaking rules; conflating NE count with SPE count.

15.8 High-Frequency PYQ Topics (ranked)

1. **Mixed-strategy NE** (drug-dealer; generic 2×2) — almost guaranteed, high marks.
2. **Sequential tree** (A/B–C/D–E/F): counting + SPE + NE count.
3. **Dominance / IESDS** (discrete auction): strict vs. weak, mixtures.
4. **Duopoly** (Cournot or Bertrand): reaction function + equilibrium.
5. **Anti-coordination** (Hawk–Dove): which outcomes can't be NE.
6. **Congestion / Braess routing**: equilibrium time, paradox after a new link.
7. **Numeric add-ons**: annuity, returns to scale.

Final word

You have, in this book, the complete machinery for Quiz 1: the theory built from first principles (Part I), every recurring question solved and verified (Part II), the gap-filling ideas (Part III), the cheat-sheets (Part IV), and the battle plan (Part V). Recognise the type, apply the recipe, check the deviation/indifference condition, and report exactly what is asked. That is full marks.

*End of Book 1 — Quiz 1 Preparation Guide.
Good luck. Made by Anmol Kansal with AI.*